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### WASHABLE FAUCET LEVER WITH CRANK-ACTIVATED DETERGENT DISPENSER

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#### Abstract

A washable faucet lever with crank-activated detergent dispenser having a lever cartridge having a stem operably coupled to a hand lever and a two-bar linkage configured to rotate in a linkage axis of rotation parallel to and simultaneous with a lever axis of rotation, a detergent reservoir member configured to house a liquid detergent therein and that includes a reservoir valve, an encasement member defining an enclosed region therein and an outlet, and having an outlet valve, wherein the dispenser includes a piston disposed within the enclosed region, coupled to the two-bar linkage, and operably configured to translate within the enclosed region upon rotating the hand lever in the lever axis of rotation to selectively and simultaneously displace the reservoir valve and outlet valve to permit flow of the liquid detergent through an internal channel of the reservoir member, into the enclosed region, and out through the outlet.

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## **Background/Summary**

CROSS-REFERENCE [0001] This application claims the benefit of provisional application Ser. No. 63/475,716 filed on Dec. 3, 2022 by the present inventor.

### **BACKGROUND TO INVENTION**

[0002] The taps of current bathroom faucets are usually activated by handles. Currently, handle-less faucets taps are predominantly located in restrooms of airports among other places. These handle-less faucets taps serve to significantly reduce the spread of contagious diseases that may have a devastating effect on the well-being of human beings amongst other possibilities. These faucets however are not economical, additionally such handle-less faucets have a limited lifespan when compared to lever enabled faucets.

[0003] Traditional manually operated faucets in homes are usually fitted with cartridge valves requiring a handle for their activation; consequently, everyone in a home or school or locations where faucets with handles are used is exposed to being infected by handle enabled faucets. Currently, handle enabled faucets require that a lever be turned for activation, however the spout is set apart from the handle; this is so because one is able to get water running by turning a lever handle through at most a right angle for existing handle enabled faucets.

[0004] Activating the lever can contaminate the lever handle with pathogens. After hands are washed for example, these same hands again are used to activate the lever in order to turn off the flow of water, consequently hands or other parts of the body may again be contaminated by the residual pathogens that initially were left on the lever before hand washing. Current faucets are not designed to wash the lever or dispense detergent or antibacterial solution when a faucet lever is activated.

[0005] Many designs of manually operated faucet valves exist; however, there is a need for a manually operated faucet enabled with lever washable detergent dispensing capabilities. Furthermore for reasons of economy and human wellbeing such a lever washable detergent dispensing faucet should be affordable. Additionally, such a design of a lever washer detergent dispensing faucet valve should be in keeping with creative modern faucet design features, be long lasting while being disposed to manipulation by a consumer.

[0006] Additionally, the advent of globalization facilitates the rapid spread of diseases that could significantly impact world health and economies on large scale; such diseases previously were localized and thus did not pose a threat to the global economic or well being of humanity. It is conceivable that there are unknown pathogens and that there will always be the threat of life-threatening pathogens at this point of global human development and beyond.

[0007] Electronic eye faucets or handle-less faucets are available, for example, as seen in Stauffer, U.S. Pat. No. 4,767,922, entitled a Hand Presence Activated Water Faucet Controller, or in Chen, U.S. Pat. No. 9,347,207, entitled Faucet Assembly. However, these faucets are not generally affordable by the global population in general. Consequently, a faucet system that inherently enables a showering of the lever and dispensing hand washing detergent while being affordable thus mitigates the spread of pathogens in schools, homes and other institutions and facilities where people share the use of faucets.

[0008] Furthermore, by integrating a lever activated soap dispenser with a washable lever significant reduction in the transmission of pathogens can be achieved.

### **BRIEF SUMMARY**

[0009] According to an embodiment of the disclosed subject matter, the invention provides greater protection against the transmission of pathogens such as the Covid-19 virus, H1N1 virus, flu virus and other transmittable pathogens when hands are being washed. The invention includes a lever handle enabled valve cartridge that efficiently operates while dispensing detergent and enabling its lever handle to be washed when hands are being washed.

[0010] The invention includes a stem assembly that functions to enable the lever handle to be washed by water flowing from the spout of a bathroom faucet tap utilizing preferably a valve cartridge; this is enabled by appropriately dimensioning the length of the lever in relation to the positioning of the spout and faucet cartridge of the invention thus enabling lever to intercept the downward direction of flow of water from the spout of a faucet using preferably a cartridge valve.

[0011] The stem also initiates a detergent dispensing unit to dispense detergent from a detergent dispensing unit while being enabled to reduce the flow of solvent through the spout of the faucet unit when the lever is turned to enable flow of water through the spout of a faucet. The stability of the stem is also enabled by the application of a bonnet nut that stabilizes the stem to a cartridge mount. Embodiments of the invention enable detergent to be dispensed preferably prior to substantial flow of water from a faucet spout when the lever is initially turned or during timely rotation of the lever as water flows from the spout of the faucet as the lever is turned.

[0012] The faucet assembly is disposed to have at least one valve cartridge, with the assembly comprising a faucet having at least one faucet handle; a fluid conduit influenced by the faucet handle for activating the cartridge valve, a faucet body having an outlet or spout for releasing fluid at an outlet end and an inlet end in fluid communication with the fluid conduit; a base for mounting the faucet handle and faucet body; the base having an upper surface and at least one cartridge comprising a cartridge body having an exterior surface, an interior surface defining a longitudinally extending passage extending from an inlet opening to an outlet opening, and at least one side opening extending transversely through the body from the interior surface to the exterior surface.

[0013] The stem of the cartridge unit is enabled with an upper end portion and a lower end portion. The upper end portion of the stem is enabled with means to rotate in reaction to the turning of a lever. The upper end portion is further enabled with means for initiating activation of a detergent dispensing unit and also with attachments enabling restriction the rotation of the lever to its predetermined maximum angle of rotation. The upper end portion extends outwardly beyond the outlet opening of the body when the stem member is situated in the passage of the body. The lower end portion is cylindrically configured to fit in rotatable relationship with the interior surface of the body when situated within the passage of the body. Additionally, the lower portion has a bottom surface configured to contact and interact with the upper surface of a moveable disc. The geometric side wall of the lower portion configured to fit within the passage of the body and in rotatable mirrored relationship with the interior surface of the body and with an inwardly extending sidewall portion defining at least one fluid receiving area wherein the receiving area is configured so that it can be rotated to align with the side opening in the body for allowing fluid flow out of the side opening of the body.

[0014] The fluid receiving area is somewhat parabolic in features and bordered by the surface of a fixed disc, the lower surface of the stem and one cartridge body sidewall opening.

[0015] The receiving area is configured so that it can be rotated to align with the side opening in the body for allowing fluid flow out of the side opening of the body; a stationary disk having at least one opening, but preferably two symmetrically arranged there through for allowing passage of fluid from an inlet opening of the body through the side opening of the body when the opening in the stationary disk is aligned with the fluid receiving area in the moveable disk, the stationary disk having a side wall capable of interacting with the interior surface of the body; and a sealing ring for stabilizing and retaining the stationary disk and moveable disk within the passage of the body, the sealing ring designed simultaneously capable of mating with the body and interfacing with a conduit to ensure stability and reliability for the interacting stress of its function.

[0016] The exterior surface of the body preferably comprises at least one sealing groove for receiving a sealing component such as an o-ring. The exterior surface of the body may also be enabled with means for seating the valve cartridge within or on a fluid conduit; for example, the feature may be a locking tab and the body may comprise two diametric locking tabs on. The upper surface of the cartridge body may be enabled with means for limiting rotational movement of the valve cartridge when in use; for example, there may be two raised preferably rectangular prismatic shaped stop features on a circle of the upper body surface. The position of the raised stop features a function to restrict the rotation of the lever to its maximum predetermined angle of rotation. The lower end portion of the stem may have a groove on the exterior for receiving a seal, such as an o-ring, between the interior surface of the body and the lower end portion of the stem. The interior surface of the cartridge body may also be enabled with a groove near the inlet end of the body for receiving a sealing ring such as an o-ring. The sealing rings may preferably be configured from suitable flexible and elastomeric material such as silicone or nitrile rubber or appropriately compatible allied sealing intervention such as fluoro-elastomers.

[0017] The stationary disc and moveable disc are preferably configured from ceramic. The stem preferably is made from metal, preferably stainless steel or brass. The upper end cartridge body is preferably made from copper, brass, or stainless steel.

[0018] The cartridge body preferably has two side openings, and the stationary disk preferably has two openings for allowing passage of water from the inlet opening of the body then through the side openings of the body when the openings in the stationary disk are aligned with the fluid receiving area in the moveable disk. The cartridge body is preferably made from copper, brass, stainless steel or polymeric or hard rubber composite such as polyether ketone, polyphenylene sulfides, Bakelite or allied compatible hard composite rubber or plastic.

[0019] Preferably, there are two fluid receiving areas in the moveable disk, and the stationary disk is enabled with locking mechanism that is capable of mating with a mating groove on an exterior surface of the body thus ensuring restrictive rotary motion. The preferably two opening access on the stationary disk is further oppositely aligned to accommodate the function of the lever assembly; the function of the lever assembly is to ensure that the lever can be enabled for washing while ensuring convenient location of the lever relative to the function and positioning of faucet parts when not in use.

[0020] The relational distances between the base of the faucet spout and the base of the spindle handle a, the distance between the vertical line of fluid stream from the faucet spout and the base of the spout b and the distance between the vertical line of fluid stream and the base of the faucet handle c satisfy the relationship  $\sqrt{a^2+b^2} \leq c$ .

[0021] The said washable lever handle initiate flow of solvent by activating rotation of the spindle thus a cam attached to the upper end of the stem will rotate when the lever is rotated, hence by attaching a crank and connecting rod to the lever where such a crank influences a follower with connection to a piston where the piston functions as a projection retraction activating unit for dispensing detergent, significant reduction in the transmission of pathogens can be achieved.

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## Description

### DESCRIPTION OF DRAWINGS

[0022] The accompanying drawings, which are included to provide further understanding of the disclosed subject matter, are incorporated in and constitute a part of this specification. The drawings also illustrate embodiments of the disclosed subject matter and together with the detailed description serve to explain the principles of embodiments of the disclosed subject matter. No attempt is made to show structural details in more detail than may be necessary for a fundamental understanding of the disclosed subject matter and various ways in which it may be practiced.

[0023] FIG. 1 shows an embodiment of a two-handle embodiment of the invention as seen when a finished unit of the invention is mounted and available for utilization.

[0024] FIG. 2 shows a positioning between the lever, the spout, the cartridge and the base of the spout of a finished unit of the invention. The detergent reservoir **450** and a spout escutcheon **977** are not shown in FIG. 2.

[0025] FIG. 3 shows a right-side view of that of FIG. 2.

[0026] FIG. 4 shows a geometric relation for the positioning of the lever, the spout, the cartridge and the base of the spout for a finished two handle unit of the invention.

[0027] FIG. 5 shows a geometric relation for the positioning of the lever, the spout, the cartridge and the base of the spout for a single handle unit of the invention.

[0028] FIG. 6 shows an embodiment of the cartridge unit **420**.

[0029] FIG. 7 shows an exploded view of the cartridge unit of FIG. 6.

[0030] FIG. 8 shows a view of the cartridge body **90**.

[0031] FIG. 8a shows a diametric section of the cartridge body **90** shown in FIG. 8.

[0032] FIG. 9 shows a view of a bonnet nut **205**.

[0033] FIG. 10 shows a section through the bonnet nut of FIG. 9

[0034] FIG. 11 shows a view of the net of lever assembly **304**.

[0035] FIG. 12 shows a view of the lever assembly of FIG. 11.

[0036] FIG. 13 shows a view of a stop interceptor.

[0037] FIG. 14 shows a view of the stem member **104**.

[0038] FIG. 15 shows a view of the moveable disc **102**.

[0039] FIG. 16 shows a view of the stationary disc **112**.

[0040] FIG. 17 shows a planer view of stationary disc **112**.

[0041] FIG. 18 shows the stationary disc inserted in the interior of the lower end **62** of cartridge body **90**.

[0042] FIG. 19 shows a planer view of movable disc **102**.

[0043] FIG. 20 shows an integration of the stationary disc **112** and the movable disc **102** inserted in the lower part **62** of the cartridge body **90** when in closure mode.

[0044] FIG. 21 shows an integration of the stationary disc **112**, the moveable disc **102** inserted in the lower part **62** of the cartridge body **90** when in open mode.

[0045] FIG. 22 shows a view of the cartridge from opening **82** in the closure mode.

[0046] FIG. 23 shows a section through the cartridge of FIG. 22.

[0047] FIG. 24 shows a view of the cartridge of FIG. 22 rotated counterclockwise by 90 degrees.

[0048] FIG. 25 shows a view of a section of the cartridge of FIG. 24 in a plane perpendicular to the plane of FIG. 23.

[0049] FIG. 26 shows the stationary disc **112**, orientation of the moveable disc **102** and a configuration template **122** for receiving stationary disc **112** and moveable disc **102** when the system is in closure mode.

[0050] FIG. 27 shows an integration of the stationary disc **112**, the movable disc **102**, in the configuration template when the cartridge is in closure mode.

[0051] FIG. 28 shows the stationary disc **112**, orientation of the moveable disc **102** and a configurative template **122** for receiving stationary disc **112**, moveable disc **102** when the system is in open mode.

[0052] FIG. 29 shows an integration of the stationary disc **112**, the moveable disc **102** in the configuration template **122** when the cartridge is in open mode and maximum rate of flow is enabled through the faucet spout **10**.

[0053] FIG. 30 shows an integration of the stationary disc **112**, the moveable disc **102** in the configurative template **122** when the cartridge is in open mode and the lever **20** has turned from its at rest position through its maximum predetermined angle of rotation thereby enabling in reducing rate of flow through spout **10** of the faucet by partially blocking openings **114**.

[0054] FIG. **31** shows a configuration in closure mode of the cartridge of the orientation of lever **20**, the stationary disc **112** and the moveable disc **102**.

[0055] FIG. **32** shows a configuration of lever **20**, the stationary disc **112** and the moveable disc **102** when the cartridge is in open mode and maximum rate of flow is enabled through spout **10**. Lever **20** does not abut stop **61**.

[0056] FIG. **33** shows a configuration of lever **20**, the stationary disc **112** and the moveable disc **102** when the cartridge is in open mode and the lever **20** has turned its maximum predetermined angle thereby reducing flow. Lever **20** abuts stop **61**.

[0057] FIG. **34** shows a view of FIG. **31** for a left-handed user.

[0058] FIG. **35** shows a view of FIG. **32** for a left-handed user.

[0059] FIG. **36** shows a view of FIG. **33** for a left-handed user.

[0060] FIG. **37** shows a configuration of the cartridge unit **604** in closed mode. Crank unit **410** and lever unit **304** not shown.

[0061] FIG. **38** shows a configuration of the cartridge unit **604** in open mode. Crank unit **410** and lever unit **304** not shown.

[0062] FIG. **39** shows an embodiment of a crank unit **410**.

[0063] FIG. **40** shows an embodiment of a cartridge unit **420**.

[0064] FIG. **41** shows a section view of a detergent activation unit in a vertical plane of symmetry for an embodiment of the detergent activation unit **430**.

[0065] FIG. **42** shows a section view of a valve for regulating detergent fluid from a fluid reservoir to a detergent activation unit **440**.

[0066] FIG. **43** shows a section of an embodiment of a detergent reservoir unit **450**.

[0067] FIG. **44** shows a configuration of the assembling of crank unit **410** and cartridge unit **420**, with a section view of the detergent activation unit **430**, a section view of the valve regulating unit **440** and a section view of the detergent reservoir unit **450**.

[0068] FIG. **45** shows labeled parts of crank unit **410**.

[0069] FIG. **46** shows the positioning of crank unit **410** on cartridge unit **420**. Bonnet nut **205** is not shown.

[0070] FIG. **47** shows the positioning of the crank unit **410** and lever unit **305** on cartridge unit **420**. Bonnet nut **205** is not shown.

[0071] FIG. **48** shows an external view of an embodiment of the detergent activation unit **498**.

[0072] FIG. **48a** shows a view of region **466** for FIG. **48**. FIG. **48b** shows a view of region **468** for FIG. **48**.

[0073] FIG. **49** shows the labeled parts of the encasement unit **498** for the detergent activation unit in a vertical plane of symmetry for the internal configuration of encasement unit **498**.

[0074] FIG. **50a** shows a section view through line **560** in a vertical plane at right angles to the cylindrical height of the encasement unit **498** shown in FIG. **49**.

[0075] FIG. **50b** shows a section view in a vertical plane through line **510** at right angles to the cylindrical height of encasement unit **498**.

[0076] FIG. **50c** shows a section view in a vertical plane through line **520** at right angles to the cylindrical height of the encasement unit **498**.

[0077] FIG. **50d** shows a section view in a vertical plane through line **530** at right angles to the cylindrical height of the encasement unit **498**.

[0078] FIG. **51** shows the labeled parts of detergent activation unit **430** in a vertical plane of symmetry.

[0079] FIG. **52** shows the labeled parts of detergent activation unit **430** in a horizontal plane of symmetry.

[0080] FIG. **53** shows a sectional view embodiment of a valve dispensing unit **484** in a diametric plane with **471**, **474** and **482** with spring **483** is not shown in said diametric plane.

[0081] FIG. **54** shows a section view of an embodiment end cap member **475** of the valve

dispensing unit **430**.

[0082] FIG. **55** shows an embodiment of the spring member for the valve dispensing unit **483**.

[0083] FIG. **56** shows a section view of an embodiment of a member **470** for the valve dispensing unit **484**.

[0084] FIG. **57** shows a section view in a plane symmetric to the cylindrically disposed height of an embodiment of the piston unit **449** for the detergent activation unit.

[0085] FIG. **58** shows a view through line **500** of FIG. **56** in a plane at right angles to the cylindrically disposed height of member **485**.

[0086] FIG. **59** shows a view from the left of piston unit **449** of FIG. **57**.

[0087] FIG. **60** shows a section view through line **502** in a plane at right angles to the cylindrically disposed height of piston unit **449** for FIG. **57**.

[0088] FIG. **61** shows a section view through line **504** in a plane at right angles to the cylindrical disposed height for member **493**.

[0089] FIG. **62** shows a section view in a vertical plane of the cylindrically disposed inlet detergent valve unit **440**.

[0090] FIG. **63** shows a section view in a vertical plane of encasement **680** for inlet detergent valve unit **440**.

[0091] FIG. **64** shows a relation view in a vertical plane for members **681** and **682** for the inlet detergent valve unit.

[0092] FIG. **65** shows a spring component **621** for the inlet detergent valve unit **440**.

[0093] FIG. **66** shows a view in a horizontal plane containing the line **700** of FIG. **67** looking up to **681**.

[0094] FIG. **67** shows a view in a horizontal plane containing the line **702** of FIG. **67** looking down to **682**.

[0095] FIG. **68** shows a view in a vertical plane of symmetry for members **682** and **632** of the inlet valve unit **440**.

[0096] FIG. **69** shows a section outline view in a horizontal plane containing line **704** looking down to **682**.

[0097] FIG. **70** shows a view in a vertical plane of symmetry for the internal parts of the valve unit of **440**.

[0098] FIG. **71** shows a view in a vertical plane of the configuration of the detergent reservoir unit **450**.

[0099] FIG. **72** shows a view in a vertical plane of symmetry of a choke point for enabling flow or disabling of detergent from the detergent reservoir.

[0100] FIG. **73** shows a view of a stopper for enabling or disabling flow of detergent through choke point **720**.

[0101] FIG. **74** shows a top view of the detergent reservoir unit **450**.

[0102] FIG. **75** shows a section in a vertical plane of the stopper **726** inserted into the choke point **720** in the closure mode.

[0103] FIG. **76** shows a section in a vertical plane of the stopper **726** inserted into the choke point **720** in the open mode.

[0104] FIG. **77** shows a vertical section through the detergent unit and the detergent activation unit when assembled and the invention is in the off mode for the position of the lever.

[0105] FIG. **78** shows a section in a horizontal plane through the detergent activation unit when assembled and the invention is in the off mode. The position of the lever with atop view of the crank unit **410** is also shown.

[0106] FIG. **78a** shows a phase of configuration of the detergent activation unit in the closed mode for showing a relation between the input of detergent valve **632**, the piston unit and the output detergent valve **482**.

[0107] FIG. **79** shows a vertical section through the detergent unit and the detergent activation unit

when assembled and the invention is in an activation mode for the position of the lever.

[0108] FIG. **80** shows a section in a horizontal plane through the detergent activation unit when assembled and the invention is in the activation mode for the position of the lever with top view of the crank unit **410**.

[0109] FIG. **80a** shows a phase of configuration of the detergent activation unit in the activation mode for relating a relation between the input of detergent and the output of detergent. The input valve **632** is closed while the output valve **482** is open.

[0110] FIG. **80b** shows position of the piston as related to the opening of valve outlet valve **484** and the closing of valve **632**.

[0111] FIG. **81** shows a vertical section through the detergent unit and the detergent activation unit when assembled and the invention is in the activation mode for the maximum predetermined rotation of the lever. The piston unit **491** would have retracted.

[0112] FIG. **82** shows a section in a horizontal plane through the detergent activation unit when assembled and the invention is in the activation mode for the position of the lever in a top view.

[0113] FIG. **82a** shows a phase of configuration of the detergent activation unit in the activation mode for a relation between the input of detergent and the output of detergent and the closing of the detergent inlet valve and the closing of the detergent outlet valve.

[0114] FIG. **82b** depicts a configuration of the detergent activation unit **430** for closure of outlet valve **484** and the opening of inlet valve **632**.

[0115] FIG. **83** shows a vertical section in a plane symmetric to the detergent activation unit and the detergent reservoir in relation to the position of the crank unit **410**, the cartridge unit **420** and the lever unit **304** in the inactive or closed mode of the invention.

[0116] FIG. **84** shows a configuration of the lever, the moveable disc **102** and the fixed disc **112** for the closed mode for FIG. **88**.

[0117] FIG. **85** shows a vertical section in a plane symmetric to the detergent activation unit **430** and the detergent reservoir unit **450** in relation to the position of the crank unit **410**, the cartridge unit **420** and the lever unit **304** in the active or open mode of the invention.

[0118] FIG. **86** show a configuration of lever **20**, the moveable disc **102** and the fixed disc **112** in the open mode for FIG. **90**.

[0119] FIG. **87** shows configuration **820** for mounting and stabilizing the cartridge unit **420** for use with a wash basin.

[0120] FIG. **88a** shows a configuration **861** for interfacing with the cartridge unit **420** and enabling water from an inlet to the cartridge valve **420**.

[0121] FIG. **88b** shows configuration **863** for directing water from an inlet to the interior passage of **861**.

[0122] FIG. **88c** shows a section through **963** of **861** shown in FIG. **88b**.

[0123] FIG. **88d** shows a configuration for directing water from **829** of outlet cartridge mount **820** to a conduit connected to spout **10**.

[0124] FIG. **88e** shows a section of the assembly of the cartridge **420**, bonnet nut **205**, and inlet interfacing member **861** to the cartridge mounting **820**.

[0125] FIG. **89** shows a section of the assembly of the cartridge **420**, bonnet nut **205**, inlet interfacing member **861**, cartridge mounting and stabilizing member **820** and member **865** and inlet member **869**.

[0126] FIG. **90** shows a configuration of the mounting block **981** for the invention in relation to a wash basin.

[0127] FIG. **91** shows a configuration of an under-cover plate for the invention when mounted in relation to a wash basin.

[0128] FIG. **92** shows a configuration of an escutcheon for the invention when mounted with a wash basin.

[0129] FIG. **92a** shows a configuration of the spout assembly.

[0130] FIG. **93** shows the first step for mounting and securely attaching the invention to its mounting base with rubber washers **834** and nuts **833**. The detergent activation unit is also mounted to the mounting base.

[0131] FIG. **93a** shows a front view of the first step for mounting and securely attaching the invention to its mounting base with rubber washers **834** and nuts **833**.

[0132] FIG. **94** shows the next phase in the assembling the invention with the cartridge unit **420**, then the turning stop restrictor **404**, then the bonnet nut **205** and then the cam unit **410**.

[0133] FIG. **94a** the escutcheon **940** is next assembled.

[0134] FIG. **94b** shows the assembly of spout unit **975**, the pop-up unit and lever unit **304**.

[0135] FIG. **95** shows an assembly of the invention without the detergent reservoir unit **450**.

[0136] FIG. **96** shows a configuration of the assembly of the invention when mounted with a wash basin and the detergent reservoir unit **450**.

[0137] FIG. **97** shows and exploded view of the configuration of the assembly of the invention.

[0138] FIG. **98a** shows an image of the configuration **861'** for interfacing with the cartridge unit **420** and enabling water from an inlet to the cartridge valve.

[0139] FIG. **98b** shows an alternative configuration **863'** with compatible with means for connection to **861'** interface, a cartridge mount **820'**, an input conduit to interface **861'** and an output from a cartridge mount **820'**.

[0140] FIG. **98c** shows a section of an alternative configuration **820'** for the mounting of the cartridge unit **420**, bonnet nut **205**, and inlet interfacing member **861'**.

[0141] FIG. **99** shows a section of the assembly of the cartridge unit **420**, bonnet nut **205**, inlet interfacing member **861'**, cartridge mounting and stabilizing member **820'** and member **863'**.

[0142] FIG. **100** shows an alternative configurative **104'** for the stem unit of the invention.

[0143] FIG. **101** shows an alternative configuration **89'** for the stop interception.

[0144] FIG. **102** shows the cartridge unit assembled with alternative stop **89** and alternative stem **104'**.

[0145] FIG. **103** shows an alternative configuration for the lever handle unit for a rectangular cross-section inlet for the cartridge stem.

[0146] FIG. **104** shows a section of the bonnet nut with a circular cross-section for stabilizing the alternative cartridge system.

[0147] FIG. **105** shows a view of the bonnet nut **205** with a circular cross-section **205** for stabilizing the alternative cartridge system.

[0148] FIG. **106** shows a view of the crank unit **810** with a rectangular cross-section for assembly with the stem alternative **104'**.

[0149] FIG. **107** shows an alternative configuration for the cartridge unit assembled with stop **89** and crank alternative **810**.

[0150] FIG. **108** shows a cam alternative embodiment **410''** for activating a detergent activation unit.

[0151] FIG. **109** shows said cam alternative **410''** assembled with the cartridge valve unit **420**.

[0152] FIG. **110** shows the cartridge valve unit assembled with alternative cam embodiment **410''** and lever **20**.

#### DETAILED DESCRIPTION

[0153] Referring to FIG. **1**, what is shown is configuration of an embodiment of the invention as seen in a faucet mount in relation to a wash basin.

[0154] Referring to FIG. **2** what is shown a geometric positioning relation of the base **12** of a faucet spout **10**, and distance **14**, labeled a of said base **12** of said faucet spout **10** from said base **12** of spout **10** to a base **16** of a lever handle **20** of said faucet where said lever handle **20** is enabled with means for rotational motion when said lever handle **20** is attached to cartridge assembly **420**. Said lever handle **20** rotation being enabled by connection to the upper portion of a spindle of a faucet cartridge valve. Said lever **20** is preferably configured with an upper flat surface region **201** and a

lower curved surface region **202** to better to facilitate disbursement of liquid on said upper flat surface region **201** and adhesion of liquid to said lower curved surface region **202** thus substantially facilitating washing of total surface area of said lever **20** when said lever **20** is disposed to intercept line of flow **777** from spout portion **8**. The members detergent reservoir **450** and spout escutcheon **977** are not shown in FIG. 2. Said lever **20** is preferably made from brass or stainless steel.

[0155] Referring to FIG. 3, what is shown is a view from the right of FIG. 2. Line **777** shows a direction of vertical flow of fluid from a faucet spout portion **8** of said spout **10** on said faucet mount. The horizontal distance **24**, labeled b, between line **444** and line **777** and in the plane containing line **444** and line **777** is such that the vertical line **555** projects between line **444** and line **777** and in the plane containing line **444** and line **777** and where line **555** is the vertical line of axis **555** of said cartridge unit **420**.

[0156] Referring to FIG. 4 and FIG. 2 geometric length **26** labeled c of said handle lever **20** can be related to the said geometric distance a and said geometric distance b by the inequality  $\sqrt{a^2 + b^2} < c$ . When said relation  $\sqrt{a^2 + b^2} \leq c$  exists in said faucet said lever handle **20** can be enabled to intersect said line of fluid flow **777** from said spout portion **8** of spout **10** of said faucet; thus said lever portion **20** can be exposed to interact with said fluid flowing from said spout portion **8** of said spout **10**.

[0157] Referring to FIG. 5, what is shown is a geometric configuration for a single lever faucet unit.

[0158] Referring to FIG. 6, what is shown is an embodiment of a cartridge unit **420**. Referring to FIG. 7 what is shown are the exploded components of said cartridge unit **420**. As shown in FIG. 6 cartridge **420**, includes a body **90**, said body **90** has an exterior surface **64** and an interior passage **68**. FIG. 8 shows a configuration of said body **90** while FIG. 8a shows the configuration of a cross-section of said body **90**. As shown in FIG. 8 the exterior surface is preferably generally cylindrical in shape so as to be easily situated in a flow conduit for fluid and thus enable the action of an inline valve between the conduit and the outlet of a faucet. The body may be formed of a variety of materials such as metal, metal alloys or other utility compatible economical nonmetallic composite or thermoplastic.

[0159] The interior passage **68** of the body **90** defines a longitudinally extending passage **68** through the body as shown in FIG. 8a. Passage **68** preferably extends from an inlet opening **70**, at the bottom of the body to a lateral outlet opening **82** at the top of the body. Passage **68** preferably has a cylindrical configuration conducive to use of rotating parts therein and can also have one or more sections or ones to accommodate the size and shape of various parts therein. As shown the passage has an upper portion **76** which is preferably of a narrower cross-sectional diameter as shown in FIG. 8a to correspond to the dimensions of the region **57** of said stem **104**, and a lower wider cross-sectional diameter section **78** that is preferably wider to accommodate other parts of the valve such as a disk, sealing rings and rotatable base portion of said stem **104**.

[0160] The body further includes at least one side opening **82** extending transversely through the body from the interior passage **68** to the exterior surface of the body **90**. The side opening **82** is preferably located to be in keeping with current faucet design in that water or other fluid flowing upwards through the inlet end **70** of the body into passage **68** would encounter the stationary disk and pass through the opening therein when in the open valve mode until it entered the receiving area of the moveable disk and could then flow out the side opening **82** in the body to continue on to feed the faucet body and outlet of the faucet.

[0161] Thus, one or more side opening **82** for this embodiment should preferably be positioned to accommodate that flow out said side openings **82** in an unimpeded manner within the design. The opening is preferably rectangular for fluid flow characteristics but can have other configurations. As shown two such side openings **82** are shown which is compatible with the number of openings and receiving areas in the disk cartridge.

[0162] The openings **82** is preferably rectangular for fluid flow characteristics but can have other

configurations. As shown two such side openings **82** are shown which is compatible with the number of openings and receiving areas in the disk cartridge.

[0163] FIG. **9** shows a configuration of body stop interceptor member **205** while FIG. **10** shows a cross-section of said member **205**.

[0164] Referring to FIG. **8**, said body **90** has an upper substantially flat surface **130** enabled with means for limiting the rotation of handle **20** and thus the rotation of stem **104**. Said means for limiting the rotation of stem **104** is enabled by raised feature **60** and raised feature **61** where said feature **60** and said feature **61** are designed to limit the turning of said stem **104** by said lever **20** to a maximum predetermined angle of rotation; such limiting is enabled when said stop **60** and said stop **61** are rigidly connected to said upper surface **130** of said cartridge body **90**. As will be shown said maximum predetermined angle of turn limits the range of motion of a moveable disc **102** against the openings of a fixed disc **112** and thus enables desired coordination of the various flow openings within a particular predetermined rotational range. The embodiment of such a moveable disc is shown in FIG. **15** while the embodiment of such a fixed disc is shown in FIG. **16**. Said stop feature **60** and said stop feature **61** are enabled on said upper surface **130** and arranged such that with the centre of a circle lying on the axis of rotation of stem portion **57**, said stop features location **60** and **61** may characterize end points on separate radii where said radii originate from a common point lying in the plane of surface **130** on the axis of rotation of stem portion **57**.

[0165] Referring to FIG. **10**, interior flange **47** of bonnet nut **205** shown in FIG. **9** contacts outer region of body surface portion **130**. Access region opening **45** of said bonnet nut **205** mates with and surrounds region **59** of stem **104**; however, said access opening **45** of said bonnet nut **205** is enabled to rotate substantially freely around region **59** of stem **104**. Said bonnet nut **205** fully accommodates passive stop **60**, passive stop **61** and said active stop **404** such that a measurable distance exists between the interior region of **205**, passive stop **60**, passive stop **61** and active stop **404** thus enabling stability of stem **104** when said active stop **404** is mated to region **56** of said stem **104**. Member **65** of active stop **404** interact with static stop **60** and static stop **61** to substantially enable the maximum predetermined angle of rotation of stem **104**. The bonnet nut **205** functions to ensure stability of the stem **104** and by extension, the stable function of the cartridge unit **604**.

[0166] Referring to FIG. **14**, what is shown is cartridge unit member **104** generally called the stem of a ceramic faucet unit. Said stem **104** includes a reduced cylindrical diameter region **57** of said stem member **104**. The upper region **58** of said region **57** is contiguous with a lower region **52** of said region **57**. Said region **58** is configured to be in rotatable relationship with lever handle unit **304** of FIG. **12** while said region **52** will be shown to be configured to be in rotatable relation with faucet system member **410** shown in FIG. **39**.

[0167] Contiguous to region **52** is region **59** of said region **57**. Said region **59** is configured to be substantially frictionless and configured to move substantially frictionless against the interior **45** of component member **205** shown in FIG. **10**.

[0168] Contiguous to region **59** of said region **57** is region **56** of said region **57**. Said region **56** of FIG. **14** is configured to be in rotatable relation with interior **34** of **404** depicted in FIG. **13**.

[0169] Referring to FIG. **13**, faucet unit member **404** when assembled with region **56** of upper stem portion **57** functions to interact with body stops **60** and **61**. Said member **65** of member **404** will be shown to interact with body stop **61** and body stop **60** and also will be shown to enable a predetermined angular rotation of region **57** and consequently, ensure rotation of a moveable disc **102** by said stem **104**.

[0170] Referring to FIG. **12** portion **58** of stem **104** is enables rotation of said stem **104** when said stem **104** is in locking relationship with lever unit **304**; such locking means between exterior region **58** of region **57** of **104** and interior **34** of lever unit **304** may be enabled by a grooved locking between region **58** of portion **57** and the interior wall of circular slots **34** of said **304**. The interior wall **34** of **410** is also grooved for locking means with region **52** of region **57** of said stem **104** while the interior **34** of **404** is grooved for locking means with region **55** of region **57** of said stem

**104.** Region **59** of region **57** is substantially smooth and enables rotation of the interior surface **45** of **205** to substantially frictionless motion on the said surface **59** of region **57**. FIG. **9** shows a configuration of member **205** while FIG. **10** shows a cross-section of said member **205**. It will be shown that such locking may be enabled by rectangular slots.

[0171] As shown in FIG. **14** region **57** is part of the stem **104** of said cartridge unit **604**. Said region **57** has an upper portion **58** configured to fit in rotatable relationship with the interior surface **34** of lever assembly **304** shown in FIG. **12**, and with the lower portion **66**, configured to be accommodated within the body **90** when said stem **104** is situated within the passage **68** of said body **90**.

[0172] As shown in FIG. **14** region **57** is part of stem **104** of said cartridge unit **604**. Said region **57** has an upper portion **52** contiguous to upper region **58** configured to fit in rotatable relationship with the interior surface **34** of crank assembly **410** shown in FIG. **39**, and with the lower portion **66**, configured to be accommodated within of the body **90** when said stem **104** situated within the passage **68** of the body **90**.

[0173] As shown in FIG. **14** the stem **104** has a region **59** configured to rotate independently and substantially frictionless within member **45** of bonnet nut **205** shown in FIG. **9** and FIG. **10** and with the lower portion **66**, configured to be accommodated within the interior of the body **90** when said stem **104** is situated within the passage **68** of said body **90**. Said region **59** is contiguous to member region **52**.

[0174] As shown in FIG. **14** said region **57** of stem **104** has portion **56** configured to fit in rotatable relationship with the interior surface **34** of stop assembly **404** shown in FIG. **13**, and with the lower portion **66**, configured to be accommodated within the interior of the body **90** when said stem **104** is situated within the passage **68** of the body **90**. Said region **56** is contiguous with region **59**.

[0175] Referring to FIG. **14** the stem **104** of cartridge **604** has a lower portion **66** which is configured to fit in rotatable relationship with the interior passage **68** of the body **90** when said stem **104** situated within the said passage **68** of said body **90**. As shown in FIG. **14** the upper portion **55** of said stem **104** is enabled with groove **85** for receiving at least one sealing ring **72**. Referring to FIG. **14**, when said stem **104** is integrated within passage **68** of the body **90** said sealing ring **72** functions to prevent leakage around the stem in said region **55**. As shown in FIG. **14** lower portion **66** of said stem **104** is enabled with a groove **87** at least one sealing ring **74**. Referring to FIG. **14** when said stem **104** is integrated within passage **68** of the of said body **90** said sealing ring **74** functions to prevent leakage between the interior surface of the said body **90** and the lower end portion **66** of said stem **104**. Said sealing ring **74** sits in inclusion **87** of said stem **104**.

[0176] The upper end region **57** of the stem **104** may preferably have a threaded or other snap-fit, slide in, friction fit or gripping surface for attaching the cartridge to a faucet handle so as to be rotatable and function with the handle **20**. The lower end portion **66** of the stem **104** has a bottom surface **96** to which is attached a feature **98**. The upper end portion **58** of the said region **57** as shown in FIG. **6** is configured to be fitted within a faucet handle assembly **304** shown in FIG. **12**. The stem **104** may be formed of a one-piece construction from stainless steel, brass or zinc alloy or copper alloy or other safe metal plastic.

[0177] The bottom surface **96** of said stem **104** preferably has rectangular prismatic feature **98**, extending from said surface **96**. Said rectangular prismatic feature **98** when within tolerance is accommodated within rectangular prismatic slot **100** of moveable disc **102** will be shown to allow for interaction with said moveable disc **102** so that when said upper portion of spindle **58** region **57** of said stem **104** is fitted within a faucet handle assembly **304** and lever **20** is turned a predetermined angle, it will then in turn rotate the said moveable disc **102** substantially that of said predetermined angle of said lever assembly **304** and consequently activate the open/closure feature of the cartridge valve. FIG. **12** shows a configuration of a faucet handle assembly **304**. FIG. **15** shows a configuration of a moveable disc.

[0178] Referring to FIG. 14, a region defined by 59 of region 57 moves freely substantially frictionless within reduced diameter region 45 of bonnet nut 205. Said bonnet nut functions to substantially ensure stability of the cartridge unit.

[0179] Referring to FIG. 15 what is shown is a configuration of the moveable disc 102. Said moveable disc 102 is enabled with an inclusion 110, said inclusion 110 is contained by parabolic arc 91, and mirrored parabolic arc 94, parabolic arc 92 and mirrored parabolic arc 95, line segment 96 and mirrored line segment 97. Said inclusion is further defined by line segments 95 and line segment 99. Said inclusion 110 is mirrored on an opposite surface of said moveable disc 102. Said moveable disc 102 is further enabled with a preferably prismatic inclusion 100 where said inclusion 100 functions to tolerably accept extended tab 98 of the stem 57 thus enabling turning of moveable disc 102 against fixed disc 112 when said unit 57 is activated to rotate by activation of lever 20 for its utility and disc 102 enables fluid through 114 of fixed disc 112 into harbor 110 of said moveable disc 102.

[0180] Additionally, the upper surface of said moveable disc 102 is disposed to be described as a section of a cross-section of a cylinder defined by two parallel chords shaped to fit within a generally cylindrical portion of the passage 68 in the lower portion 78 of said passage 68. Said diametric inclusions 110 in the lower portion of moveable disc 102 function as fluid receiving harbors in the said moveable disc 102.

[0181] The movable disk 102 included within the body in passage 68 is enabled with means on its upper surface 104 configured to contact and interact with the bottom surface 96 of the lower end portion of the stem 66 in a rotatable relation. As shown, upper surface 104 is preferably configured with a slot 100 for receiving a tab 98 therein. The disk upper surface may preferably be defined as a portion of the cross-section of a solid cylinder defined by symmetrically parallel chords shaped to fit within a preferred generally cylindrical portion of the passage 68 in the lower portion 78 in the passage 68. There are preferably two side wall portions defining preferably two fluid receiving areas 110 within the moveable disk 102. The receiving area 100 is configured so that when the disk 102 is rotated to align the receiving area 100 with the side opening 82 in the body 68 it will allow fluid flow out of the side opening of the body for feeding the faucet to which is installed with fluid entering the cartridge from a conduit through the open inlet 70 of the cartridge.

[0182] Referring to FIG. 16 what is shown is a stationary disk 112. Said disk 112 is fixed in position for interacting with moveable disc 102. Said fixed disc 112 is preferably provided with openings 114 extending through the stationary disk 112 for enabling passage of fluid flow from an inlet end 70 of the body 90 then into the passage 68, and through the stationary disk and then into the receiving area 110 of the moveable disk, and then out the side opening 82 of the body when the cartridge is in the open position mode. The stationary disk 112 is preferably enabled with lubricant containing pits 115; said lubricant in lubricant containing pits function to reduce friction between moveable disc 102 and stationary disc 112 thus further prolonging functional integrity of said moveable disc 102 and said stationary disc 112. Said fixed disc is further provided with diametric tabs 118; said tabs 118 enables the said disc 112 to be fixed in recesses in the interior passage 68 of the body cartridge 90.

[0183] Referring to FIG. 17 what is shown is another view of the stationary disc 112. Referring to FIG. 18, what is shown is a configuration of said fixed disc 112 positioned with diametric tabs 118 recessed in said body 90 of the cartridge unit. Referring to FIG. 19 what is shown is a view of the moveable disc 102. Referring to FIG. 20 what is shown is a relationship between said moveable disc 102 and said stationary disc 112 when cartridge valve unit 604 is in closure mode thus substantially resisting the flow of liquid from conduit 70 through passage 68 and to side openings 82. Referring to FIG. 21 what is shown is a relationship between said moveable disc 102 and said stationary disc 112 when said cartridge valve unit 420 is the open mode thus enabling the flow of liquid from conduit 70, through the passage 68 and to side openings 82. As will be shown said disabling and enabling of the flow of liquid through the valve cartridge unit is enabled when a lever

handle **20**, when fitted to portion **58** of the region **57** of the stem **104** shown in FIG. **14** can be turned to reduce and shut off flow. The handle **20** thus initiates the closing of the valve from the open mode of said valves and the opening of the valves from a closure mode of said valves when said handle **20** is initiated to turn.

[0184] Referring to FIG. **22** what is shown is a view from the side opening **82** of said cartridge unit **420** when said cartridge unit **420** is in closure mode. FIG. **23** shows a diametric cross-section of the view of FIG. **22** and here said diametric cross-section lies in a plane of symmetry across the most substantial dimensions of preferably rectangular prismatic slot **100** of said moveable disc **102**.

[0185] Referring to FIG. **24**, what is shown is a view of said cartridge **420** of FIG. **22** rotated through 90 degrees. FIG. **25** shows a diametric cross-section of the view of FIG. **24** where said diametric cross-section lies in a plane of symmetry across the lesser dimensions of preferably rectangular prismatic slot **100** of said movable disc **102**.

[0186] Referring to a cartridge configuration say FIG. **23** for clarity; for the washing of the lever **20** the turn of stem **104** when lever handle **20** is attached to said stem portion **58** of stem **104** enables opening and closing of valves **102** and **112** and thus enables utility of the invention. The washing of the lever **20** of the invention is enabled when said lever arm **20** is attached to stem region **58** of said stem **104** and said lever **20** is initiated to rotate through a maximum predetermined angle that is greater than the angle turned to enable said lever **20** to intercept the said line of flow **777** from spout **10**. The initial maximum predetermined turn of the stem **104** by lever **20** from an off mode of the cartridge **420** will be shown to reduce flow rate through spout **10** to enable spreading of detergent over ones hand with the option for washing of hands; a subsequent counter turn of said lever **20** through a predetermined angle enables increase in the flow rate through spout **10** for washing of hands using a greater flow rate; a subsequent counter turn through a predetermined angle enables said lever **20** to intercept line of flow **22** from spout **10** and also increases flow rate through spout **10** to its predesigned maximum thus enabling the choice of washing of said lever **20** and further washing of hands. A subsequent counter turn of said lever **20** through a predetermined angle enable repositioning of lever arm **20** back to a rest positioning in which flow of liquid is disabled. The reduction in flow rate of solvent through spout **10** thus enables significant conservation in the use of water as a solvent and the washing of said lever **20** significantly reduces the spread of pathogens thus enhancing health and safety.

[0187] Referring to FIG. **16** said stationary disk **112** is enabled with preferably diametric paired extension **118** for locking said stationary disc **112** within the interior passage **68** of the cartridge body **90**; thus, enabling reliability and stability of said cartridge unit **420**. As shown in FIG. **18**, said means for locking said stationary disc **112** may be achieved when inclusion is designed within the interior passage **68** of said body **90** and were said inclusions in the passage **68** of said body **90** are dimensioned to accommodate and fit said diametric extensions **118** of said stationary disc **112**. Said stationary disc **112** interfaces with the moveable disc **102** located in the passage **68** of the cartridge body **90** and with conduit **70** where preferably openings **114** in said stationary disc **112** accommodate flow to the receiving are **110** of moveable disc **102** and then through side openings **82** to the exterior of body **90** of cartridge unit **420**.

[0188] Referring to FIG. **25** said conduit **70** is external to said interior passage **68**, thus, to ensure reliability of the cartridge unit an elastomeric seal **88** is introduced to interface said stationary disc **112** and said conduit **70** in region **132** of the cartridge body **90**. Said sealing ring **88** included within component **132** functions to apply pressure between **132** stationary valve member **112**, and moveable valve member **102**. The pressure exerted by said scaling ring **88** ensures optimal functioning of the cartridge for the flow of liquid. The stationary disc **112** is also enabled with a pit like lubricant containing feature **115**; lubricant in said pit like feature **115** function to for ensure substantially frictionless movement of said moveable disc **102** over the surface of fixed disc **112** and also to prolong the functional integrity of the cartridge unit **420**.

[0189] Referring to FIG. **26**; what is shown is an explosion of the order of contiguous assembly of

the position of tabs **118**, the orientation of the moveable disk **102** and the orientation stationary disk **112** and a cross-section through the lower part **132** of the said cartridge body **90** when the faucet system shown in FIG. **1** is in the off mode but disposed to enable flow of liquid to said faucet when activated. Referring to FIG. **27** in contrast to FIG. **26**, what is shown is an integration and relative contiguous orientation of said moveable disk **102** and said stationary disk **112** in the passage **68** of the lower part **132** of said body **90** when the faucet system of FIG. **1** is in the off mode but disposed to enable flow of liquid to said faucet when activated.

[0190] Referring to **28**; what is shown is an explosion of the order of contiguous assembly of the position of tabs **118**, the orientation of the moveable disk **102**, the orientation of stationary disk **112** and a cross-section through the lower part **132** of the said cartridge body **90** when the faucet system shown in FIG. **1** is in the activated mode but disposed to enable substantial flow of liquid through spout **10** when said lever **20** does not intercept the line of flow **777** from spout **10** to a wash basin. Referring to FIG. **29** in contrast to FIG. **28**, what is shown is an integration and relative contiguous orientation of said moveable disk **102** and said stationary disk **112** in the passage **68** of the lower part **132** of said body **90** when the faucet system of FIG. **1** is in the activated mode but disposed to enable substantial flow of liquid through spout **10**.

[0191] Referring to **30**, what is shown is an integration and relative contiguous orientation of said moveable disk **102** and said stationary disk **112** in the passage **68** of the lower part **132** of said body **90** when the faucet system of FIG. **1** is in the activated mode but disposed to reduce flow of liquid through spout **10** where said reduction in flow rate of liquid conserves water as a solvent when hands are being soaped.

[0192] Referring to FIG. **31**; what is shown is the order of contiguous assembly and relative orientation of the handle **20**, the orientation of the moveable disc **102** and the orientation of the stationary disk **112** when faucet system is in the off mode and liquid flow is disabled through spout **10** of faucet assembly for a right hand side position of faucet cartridge **420**.

[0193] Referring to FIG. **32**, moveable disc **102** is shown with a line of symmetry **1020** for said moveable disc **102**; now referring to **1020** of FIG. **31** and consistent with the properties of a rigid body, said **1020** characterizes initial orientation of said line of axis of symmetry **1020** for **102** of FIG. **31** and transformation of said axis of symmetry **1020** in **102** in FIG. **32** subsequent to lever **20** executing its predetermined angle of rotation; thus indicating a sense and angle of rotation of moveable disc **102** from the inactive mode to the utility mode of configuration where said lever arm **20** is not disposed to intercept line of flow **22** from portion **8** of spout **10**.

[0194] Referring to FIG. **33** the predetermined maximum turn of moveable disc **102** is shown with a line of symmetry **1020** for said moveable disc **102**; now referring to **1020** of FIG. **32** and consistent with the properties of a rigid body, said **1020** characterizes initial orientation of said line of axis of symmetry **1020** for **102** of FIG. **32** and transformation of said axis of symmetry **1020** in **102** in FIG. **33** subsequent to lever **20** executing its predetermined maximum angle of rotation; thus when the cartridge system **604** exists in this mode the flow rate of solvent say water is significantly reduced for enabling soaping of hands and thus as mentioned enabling economic utilization of water and by extension economic, environmental and health conservation because the lever is now washed by hands that have been washed before the cartridge system is returned to its inactive mode by the turning of said lever **20** and thus also mitigating the spread of pathogens. The washing of lever **20** is enabled when said lever **20** is counter turned to intercept the line of flow **22** of liquid from spout **10** thus said lever **20** is disabled from the line of flow of solvent from spout **10** when hands are being washed.

[0195] Referring to FIG. **34**, FIG. **35** and FIG. **36** what is shown is a configuration of FIG. **31** FIG. **32** and FIG. **33** for an embodiment of the invention in which the said cartridge valve **420** is configured for activation by turning a lever counterclockwise for said systems activation. Such a system is a configuration in which the cartridge unit is left side of the base of a faucet basin and more suited for a left-handed person.

[0196] Referring to FIG. 37, what is shown is an embodiment of a right-hand side cartridge valve **420** in the inactive mode where said member **65** attached to said stem **104** would abut stop member **60** to effect closing of openings **114** of stationary disc **112**.

[0197] Referring to FIG. 38 what is shown is an embodiment of a right hand side cartridge valve **420** in the active mode and where rotation of the stem **104** through a maximum predetermined angle from the off configuration of the cartridge **420** would have been limited by stop member **61** when said member **65** of said stem **104** intercepts body stop member **61**.

[0198] Referring to FIG. 39, what is shown is an embodiment of the crank attachment component unit **410**. FIG. 40 shows an embodiment of the cartridge unit **420**, bonnet nut **205** is not shown. FIG. 41 shows a substantially symmetric section in a vertical plane of an embodiment of a detergent activation unit **430**. FIG. 42 shows a substantially symmetric section in a vertical plane of an embodiment of a valve component unit **440**. FIG. 43 shows a substantially symmetric section in a vertical plane of an embodiment of a detergent reservoir unit **450**. FIG. 44 shows an integration of components units **410**, **420**, **430**, **440** and **450**.

[0199] Referring to FIG. 45, crank component unit **410** shows a crank region **460** with a connecting rod **496**. Said crank unit **410** has a cylindrically inner grooved attachment region **462**. Said inner grooved attachment **462** locks to region **52** of the upper stem portion **57** of stem **104** in a complete unit for the invention such that rotation of stem **104** substantially simultaneously rotates said crank component unit **410**. Said crank unit component **410** is preferably made from non-corrosive metal such as copper, brass in order to ensure maximum and prolonged functional reliability.

[0200] Referring to FIG. 46, what is shown is an integration of said crank attachment component unit **410** to portion **52** of upper portion **57** for the said stem unit **104** of cartridge unit **420**. Said crank component unit **410**, said lever **20** and said upper portion **57** constitute a rigid body thus consistent with the properties of a rigid body said crank **460** will rotate consistent with the sense of clockwise or counterclockwise rotation of said lever **20** and additionally said crank **460** rotation occurs with substantially relative invariance in angular configuration relative to said lever **20**. The vertex of the angle characterizing said angular separation lies on a common axis of rotation of said stem **104** and said lever **20** consequently, enabling substantial invariance in a predetermined said angular separation of said crank component **460** and said lever **20** when said crank **460** and said lever **20** are attached to said stem **104**. Said crank **460** is thus substantially constrained to move in a plane. Additionally, said crank **460** and said lever **20** are disposed to be separately mated to said upper stem portion **57** by groove fit thus enabling ease of manipulation in positional variation in said angle of separation between said lever **20** and said crank **460**.

[0201] Additionally, a connecting rod **496**, freely hinged to said crank **460** enables translation of a piston **491** when said connecting rod **496** is attached to a piston **491** thus enabling said piston **491** to activate opening and closing of valves to enable dispensing of liquid detergent from a detergent activation unit **430**. Said another way, the connecting rod **496** and the crank **460** form a two-bar linkage. Furthermore, the function of said crank **460** and said connecting rod **496** is enabled on multiple angular position on said stem **104** and so enables convenient positioning of the detergent activation unit for initiating dispensing of liquid detergent by the action of said crank **460**. A detergent dispensing system can therefore be conveniently positioned on a faucet mounting base for interacting within an annulus for the crank unit **410**. FIG. 46 functions to show a positioning of the crank unit **410** relative to the lever assembly **304**; in an assembly of the invention the positioning of the bonnet nut **205** would precede the positioning of the crank unit **410** to be followed by the positioning of the lever unit **305**.

[0202] Referring to FIG. 47, what is shown is an integration of crank unit component **410** and lever unit **304** to said upper portion **57** of stem unit **104** of cartridge unit **420**; thus as stated said integration of crank unit **410** and lever **20** to said upper portion **57** substantially constitutes a rigid body, consequently initiation of rotation of lever **20** initiates rotation of stem **104** and thus rotation

of crank unit **410**. FIG. **47** functions to show the positioning of the crank unit **410** relative to the lever assembly **304**.

[0203] Referring to FIG. **48**, component **498** shows an embodiment of an encasement unit or encasement member **498** for coupling with a detergent activation unit **430**. Referring to component **498**, said component **498** is disposed to be characterized as an open cylinder **464** contiguous with diametric prismatic flanges **465**. Said prismatic flanges **465** are enabled with access openings **497**; said openings **497** function as a means for attaching encasement **498** firmly to a wash basin faucet mount. Said cylindrical component **464** is further enabled with a spirally grooved opening **487**; said spirally grooved opening **487** will be shown to enable the attachment of a liquid detergent reservoir member **450**. Additionally, said cylindrically disposed region **464** is enabled with means for attaching a conduit **478**; said conduit functions to direct liquid detergent from the interior of said cylindrically disposed component **464**. Said cylindrically disposed region is further characterized by region **466** and a region **468**. Said detergent activation encasement **498** is preferably configured from Bakelite, melamine or nylon.

[0204] Referring to FIG. **48a**, what is depicted is region **468** of said cylindrically disposed member **464**. Said cylindrically disposed region **468** is internally spirally grooved and will be shown to be compatible with a closed end cap **475**.

[0205] Referring to FIG. **48b**, what is depicted is region **466** of said cylindrically disposed member **464**. Said region **466** of cylindrically disposed region **464** is enabled with guide channels **492** where said guide channels will be shown to restrict a piston **491** substantially to translation motion.

[0206] Referring to FIG. **49** what is shown is an embodiment of a vertical planar cross-section of said cylindrical disposed component **464** for encasement **498** where said planar cross-section may contain the cylindrical axis of that said cylindrically disposed **464** and that axis contains the center of a circular cross-section of that said cylinder **464**. Said cylindrically disposed **464** is also enabled with grooves for containing sealing rings **480**, **481** **489** and **490**. In an assembly of the invention said sealing rings **481**, **480**, **489** and **490** function to substantially resists leakage of detergent fluid from internal region **487** of said cylindrically disposed **464**.

[0207] Referring to FIG. **49**, said cylindrically disposed component **464** has its internal region modified for insertion, translation, stability and optimal function of a piston unit component **449**.

[0208] FIG. **50a** shows a conceptual cross-section of said **464** in a plane containing the line **560** and were said plane containing line **560** is perpendicular to the cylindrical axis of said cylindrically disposed **464**.

[0209] FIG. **50b** shows a conceptual cross-section of said **464** in a plane containing the line **510** and were said plane containing line **510** is perpendicular to the cylindrical axis of said cylindrically disposed **464**. Said cross-section defined by FIG. **50b**, in a region of a certain length and on the interior wall of cylindrically disposed **464**, is enabled with detergent fluid outlet conduit attachment **478**. Said outlet attachment is sandwiched between sealing rings **480** and sealing ring **481**. It will be shown that in an assembly of a properly functioning unit of the invention an out-valve unit **482** is excluded from travel to region of a certain length defined by line **510** and line **560** of said **464**.

[0210] FIG. **50c** shows a conceptual cross-section of said **464** in a plane containing the line **520** and were said plane containing line **520** is perpendicular to the cylindrical axis of said cylindrically disposed **464**.

[0211] FIG. **50d** shows a conceptual cross-section of said **464** in a plane containing the line **530** and were said plane containing line **530** is perpendicular to the cylindrical axis of said cylindrically disposed **464**. Diametrically paired slots **492** in a region of a certain length would contain line **530**. It will be shown that said diametric slots function to substantially resist rotation of a piston unit **449** when diametrically paired flanges **495** are enabled on said piston **491** such that said piston **491** is forced by said flanges to translate in said slots **492**. Said diametrically paired slots **492** of cylindrically disposed **464** do not extend to regions of said cylinder **464** were said slots **492** would impinge the function of o-rings **489** and **490**. Sealing rings **489** and **490** are enabled with access

within regions defined by line 520 and line 530 of the interior wall of said cylindrically disposed **464**.

[0212] Referring to FIG. 49, said cylindrically disposed component **464** is enabled with a detergent containment region **487**. In a functioning unit of the invention liquid detergent is enabled to enter said detergent region **487** after detergent fluid in said region **487** has been substantially expelled. The interior wall of region **487** of cylindrically disposed **464** is enabled with a circular flange **477**. It will be shown that said circular flange **477** function to contain an outlet valve unit **482**.

[0213] Referring to FIG. 51, what is shown is an embodiment of a planar cross-section of the detergent activation unit **430**. Said detergent activation system **430** comprises a stabilizing encasement component **498** which has already been described, an outlet valve component unit **484** shown in FIG. 53 and a piston component unit **449** shown in FIG. 57. Said piston component **449** is preferably configured from Bakelite, melamine or nylon.

[0214] Referring to FIG. 53, what is shown is an embodiment of the detergent outlet unit **484**. FIG. 54, FIG. 55 and FIG. 56, show an embodiment of the individual members of said detergent outlet **484**. Referring to FIG. 54, what is shown is an embodiment of an end cap component **475** of said detergent outlet unit **484**. Said component **475** is disposed to be characterized as a solid cylinder portion **471** contiguous with a an outer spirally grooved cylindrical annulus **474** to enable spiral insertion in compatible grooved region **468** of said cylindrical disposed **464**. Said outer spirally grooved component **474** is of a lesser radius than cylindrically disposed component **471**. An outer grooved preferably solid cylindrical region **473** of a lesser radius than **474** is contiguous with a circular face member **471** but non-contiguously included by cylindrically shaped annulus **474**. Referring to FIG. 55, what is shown is a spring **483** dimensioned to be spiraled within the grooves **472** of **473** thus enabling one end of said spring **483** to be fixed when said **475** is attached in region **468** of cylindrical disposed **464**. Components **471**, **474** and **473** are disposed to be geometrically concentric. Referring to FIG. 56, component **470** characterizes cylindrically disposed contiguously concentric with a lesser radius cylindrically disposed region **476**. Said cylindrically disposed region **476** is spirally grooved to be compatible with spring member **483**. Said grooved member **476** functions to stabilize said spring **483** and preferably fully enshrouds said member **485**. When said **475**, **483** and **470** are integrated in a functioning unit of the invention the sum total of the cylindrical heights of cylinder **473** and the height of cylinder **476** is less than that of the relaxed length of spring **483** thus enabling a compressive length of said spring **483**. It will be shown that the compressive length of said spring **483** is predetermined to be greater than the maximum translation of a piston unit **449** when said invention functions according to design. Said outlet valve **484** components may preferably configured from Bakelite, melamine or nylon while the spring unit **483** is preferably made from stainless steel for resistance to the humid environment of a faucet.

[0215] Referring to FIG. 57, what is shown is an embodiment of the piston component unit **449**. Said piston component **449** is preferably cylindrically disposed and is enabled with a region **491** where said region **491** is predisposed to move substantially frictionless against the inner wall of said cylindrically disposed **464**. FIG. 58 shows a cross-section of said **491** through a line 502 where said line 502 lies in a plane perpendicular to the central axis of cylindrically disposed **464**. Said component **449** is further enabled with a beveled upper component **488** where said upper beveled component **488** is predetermined to function as a wedge for initiating motion of a valve **632**.

[0216] Referring to FIG. 57, contiguous and concentric to said cylindrically disposed **491** and in region **487** of said cylindrically disposed **464** is component **486**. Said component **486** is of a lesser radius in comparison to the radius of **491**. Said reduced radius component **486** functions to enable detergent liquid availability in said region **487** and also to initiate dispensing of fluid detergent from said region **487**. It will be shown that said reduced radius of **486** in comparison to the radius of member **491** functions as a leading edge of the piston unit **449** for initiating contact with **482** of outlet valve unit **484** to enable detergent liquid dispensing through outlet **478** and for replenishing region **487** of **498** with detergent fluid in said region **487**. Contiguous with said component **491** and

in region **466** of cylindrically disposed **464** are flange component **493** and flange component **494**. Said flanges **493** and **494** are enabled with access slots **454** and **455**. Said access slots **454** and **455** enable hinged connection of piston unit **449** to connecting rod **496**.

[0217] Referring to FIG. **57**, said piston component **449** is diametrically enabled with raised flanges **495**. Said diametrically paired flanges are preferably vertically disposed and positioned in diametric access slots **492** in region **466** of said cylinder **464**. Said diametrically paired flanges are predetermined to function within the limits of paired access slots **492**. Said diametrically paired flanges **495** function to resist rotation in the movement of piston unit **449** and thus substantially restrict the motion of piston unit **449** to translation. Additionally, said diametric flanges **495** enable piston component to stay within the influence of sealing ring **489** and sealing ring **490**.

[0218] Referring to FIG. **58**, what is shown is a cross-section through line **500** of FIG. **53**.

Referring to FIG. **59**, what is shown is a view of piston component **449** of FIG. **57** from the left.

Referring to FIG. **60**, what is shown is a cross-section projection of component **491** of FIG. **57** in a plane containing line **502** and where that plane is perpendicular to the height of cylindrically disposed **449**. Referring to FIG. **61**, what is shown is a cross-section projection of component **491** of FIG. **57** in a plane containing line **503** and where that plane is perpendicular to the height of cylindrically disposed **449**.

[0219] Referring to FIG. **52**, what is shown is a section view of said detergent activation unit **430** in a horizontal plane where said horizontal plane contains a section view of detergent outlet conduit **478**. Access slots **497** enable firm attachment of said detergent activation unit **430** to the mounting base for a faucet such that said detergent activation unit **430** is within influence of the crank **460** and follower **496** in a functioning unit of the invention.

[0220] Referring to FIG. **62**, what is shown is an embodiment of a vertical semicircular disposed cross-section of a valve component unit **440** where said valve functions to enable flow of detergent to reservoir region **487**. Components of said valve component unit **440** are labeled **680**, **681** and **682** and **621**. FIG. **63** shows components **680**, FIG. **64** shows components **681** and **682**, while FIG. **65** shows spring components **621**. Said outlet valve **484** components may preferably be configured from Bakelite, melamine or nylon while the spring unit **483** is preferably made from stainless steel for resistance to the humid environment of a faucet. Said inlet valve **440** components may preferably be configured from Bakelite, melamine or nylon while the spring unit **621** is preferably made from stainless steel for resistance to the humid environment of a faucet.

[0221] Referring to FIG. **63**, component **680** shows a cylindrically disposed containment **624**. Said containment **624** has a region **626** internally spirally grooved for receiving external spirally grooved component **623**. Enabled in the interior of said **624** at the region distal to region **626** is a shoulder **629**. As shown in FIG. **62** said valve component **682** shown in FIG. **64** is enabled to freely rest on said shoulder **629** of the interior of said **624**. Said containment **624** is externally spirally grooved for insertion connection in the conduit component **724** of an internally grooved detergent conduit depicted in FIG. **71**. Said region of **624** distal to region **626** flares conically to reservoir region **487** of said cylinder **464**.

[0222] Referring to component **681** shown in FIG. **64**, what is shown is an annulus shaped component **623** where said component **623** is spirally grooved on its outer curved surface. Said component **623** is concentrically enabled with a circular access opening **622** where said access opening **622** functions as a conduit for detergent fluid. Contiguous and concentric to component **623** is component **625** where said component **625** is an annulus shaped. The radius of said component **625** is less than the radius of component **623** but greater than the radius of component access conduit opening **622**. Said component **625** functions as a guide for spring component **621**.

[0223] Referring to component **682** depicted in FIG. **64** said component **682** is disposed to be a composite shape characterized as a solid truncated cone **634** contiguous with a preferably solid cylindrical section **627** where the larger base radius of said solid truncated cone **634** is common to the radius of said cylinder **627**. Said preferably solid cylinder **627** is concentrically contiguous with

a circular face of an alternate cylinder **631** where said cylinder **631** is enabled with a larger radius than that of said cylinder **627**. Said cylinder **631** alternate circular face is further enabled concentrically with an alternate solid cylinder **638** where said alternate solid cylinder **638** is internally spirally grooved for insertion of externally spirally grooved extension **639** of component **632** depicted in FIG. **64**. Said externally grooved extension **639** is contiguous with said **632**. Said component **632** is disposed to be characterized as a cylinder enabled with a partial spherical component where said cylindrical section is internally spirally grooved. Said cylindrical component **631** is further enabled with conduit channels **628** extending from a circular face of said cylinder **631** to the alternate face of said cylinder **631**. Said channels **628** enable the flow of detergent to said region **487** of detergent activation unit **430**.

[0224] FIG. **66** shows a relative upward looking view of **681** through a horizontal plane at right angles to the height of cylindrical annulus **625** where said plane contains the line **700** of FIG. **64**. FIG. **67** shows a depressive view of **682** through a horizontal plane containing line **702** and where that plane is parallel to a circular face of cylinder **631**. FIG. **69** shows a depressive view of **632** with its spirally grooved extension **639** through a plane containing line **704** of FIG. **68** and where that said plane is parallel to a circular face of cylinder **631**. FIG. **70** shows a vertical section of the configuration of said component **681** and said component **682** enabled with spring member **621**. Extreme ends of said spring **621** enshrouds component **625** and component **627** such that spring **621** is enabled to be compressed when said cylinder **631** is initiated to translate component **632** from said shoulder **629** of **624**. Said shoulder functions as a support base for component **631**.

[0225] Referring to FIG. **71**; what is shown is an embodiment of the detergent component containment unit **450**. Said detergent containment unit is enabled with a preferably cylindrical detergent reservoir **716** with means **726** for enabling flow or disabling flow of detergent from said detergent reservoir **716**. Component **712** of said detergent reservoir functions as a spirally grooved cap. Removal of said spirally grooved cap **712** enables refilling of said detergent containment reservoir. Said detergent containment reservoir is also enabled with an opening **714**. Said opening **714** functions to enable detergent fluid to enter region **487** of **464** when motion of said piston **491** enables a pressure difference between said region **487** and atmospheric pressure in said detergent containment reservoir **716** and said opening **714** enables pressure in said containment reservoir **716** to be that of the atmosphere. Said detergent reservoir **450** is preferably made from polypropylene, polystyrene, polycarbonate plastic or allied compatible transparent or see through recyclable plastic. Transparency or seeing through reservoir better enable management of detergent solution for timely refilling or replacement of said detergent reservoir **450**. Due to anticipated multiple interventions of the reservoir **450** for refilling or replacement a recyclable material for the reservoir **450** is preferred thus enabling environmental, energy and material conservation.

[0226] Referring to FIG. **71**, said detergent containment unit is enabled with a preferably cylindrical conduit **722** where said conduit **722** is preferably of a lesser radius than **716** but contiguous with detergent containment reservoir **716**. Said conduit is enabled with a thicker wall cylindrical region **720** where the central concentric opening or internal channel **718** is of a lesser radius than the radius of said cylindrical disposed component **722**.

[0227] Referring to FIG. **71**, the lower region of said **722** is internally spirally grooved for accommodating inclusion of valve component **440** shown in FIG. **62**. Referring to FIG. **72** what is shown is a section of an inclusion **732** enabled in said annulus region **720** where said inclusion **732** is configured to accommodate component **726** thus enabling said **726** to initiate enabling or disabling flow of detergent from said containment reservoir **716**. Said component **726** may be configured from Bakelite, melamine, nylon or other appropriate compatible material.

[0228] Referring to FIG. **73**, what is shown is component **726**. Said component **726** is disposed to be characterized as a solid truncated cone **721** contiguous with a rectangular prism **727** where said rectangular prism **727** functions as a lever for initiating motion of said truncated cone **721**. Said truncated cone **721** is enabled with a conduit **728** where said conduit is enabled with diametric

openings on the curved surface of said truncated cone **721**.

[0229] Referring to FIG. **74** what is shown is a top view of an embodiment of the containment reservoir **716**. Referring to FIG. **75** what is shown is an orientation of said component **726** in region **732** of **720** when said component **726** disables the flow of detergent from containment reservoir **716**. Said conduit **718** is then substantially perpendicular aligned to said conduit **728** such that the flow of detergent fluid is disabled from detergent from containment reservoir **716**. Referring to FIG. **76** what is shown is an orientation of said component **726** in region **732** when said component **726** enables the flow of detergent from containment reservoir **716** of detergent containment reservoir **716**. Said conduit **718** is then substantially parallel aligned to with conduit **728** such that the flow of detergent fluid is enabled from containment reservoir **716**.

[0230] Referring FIG. **77**, a clockwise sense of rotation of said lever **20** for the invention will now be enacted. A counterclockwise sense of rotation of lever **20** is a mirrored action for a clockwise sense of rotation. A description for the function of the entity is for an embodiment in which detergent is expelled substantially at about the same time that water is enabled to flow through spout **10** for the washing of hands thus a coordination between the translation of the piston substantially parallel to the axis of cylindrically disposed **464**. The compressive length of spring **483**, the radius of the crank **460**, the travel of valve **482** prior to expulsion of detergent, the retraction of the connecting rod **496** prior to completion of the maximum predetermined angle of rotation of lever **20** are enabled and dimensioned within the maximum predetermined angle of rotation of lever **20** with the radius of the crank **460**. An embodiment of the invention in which the leading edge **486** of piston **491** is substantially close to outlet valve **482** in the inactive state is preferred for timely dispensing of detergent. The washing of the lever **20** is preferably enabled on the counter rotation of said lever **20** and subsequent to the completion of the maximum predetermined angle of rotation of said lever **20** when the crank would have retreated while the lever **20** was being turned from the off position to its maximum predetermined angle of rotation. The vertical line of flow of water from spout **10** is not intercepted by lever **20** when stops **60** and **61** restrict said lever **20** to its predetermined maximum angle of rotation. Thus, as can be seen the utility of the invention disposes the use of the invention to just washing hands with detergent prior to maximum predetermined rotation of said lever **20** when water flows through spout **10** followed by a choice to wash said lever subsequent to completion of the maximum predetermined rotation of lever **20**. Consequently, when moveable disc **102** and stationary disc **112** are configured for maximum opening of **114** for a turn of 90 degrees then a predetermined maximum angle of rotation of lever **20** is greater than 90 degrees thus stops **60** and **61** are not solely disposed to being diametrically opposite for a 90 degree orientation of lever **20** and spout **10**.

[0231] Referring to FIG. **77** what is shown is a sectional view through a vertical plane of an assembled configuration of said crank component unit **410** shown in FIG. **39**, wherein said cartridge component **420** shown in FIG. **40**, said detergent activation unit shown in FIG. **41**, said inlet valve component unit shown in FIG. **42**, and said detergent containment unit **450** shown in FIG. **43** are depicted when said configuration depicted in FIG. **77** exists in an inactive configuration.

[0232] Referring to FIG. **78** what is shown is a section in a horizontal plane of the detergent activation unit **430** in association with the stem **104** and crank unit **410** of FIG. **77**.

[0233] Referring to FIG. **77** the component **486** of piston component **491** is constrained to be in substantial proximity to valve component **482** thus spring component **483** exists in its relaxed configuration, consequently enabling closure of outlet **478** by said valve component **482**. Referring to FIG. **77**, said component **631** rests on shoulders **629** when the reservoir valve **632** is deactivated consequently enabling flow of detergent through conduits **628** to region **487** of **464** when said system is primed for initiation to function where such priming action constitutes a back-and-forth motion of said piston **491** to open and close valve component **632**. Said back and forth motion of piston component **491** initiate opening and closing of valve **632** thus enabling a pressure difference

between interior volume **716** of detergent reservoir **450** depicted in FIG. **71** and region **487** of cylinder **454** for the detergent activation unit **430** consequently enabling fluid flow of detergent to **487**.

[0234] Referring to FIG. **78** what is shown is a sectional view in a horizontal plane of an integrated configuration of said crank component unit **410** shown in FIG. **39**, said cartridge component **420** shown in FIG. **40**, said detergent activation unit shown in FIG. **41**, said inlet valve component unit shown in FIG. **42** and said detergent containment unit shown in FIG. **43** when said configuration depicted in FIG. **81** exists in an inactive configuration. Member **495** is positioned within groove **437** of crank unit **410**.

[0235] Referring to FIG. **78a**, what is shown is a diagrammatic representation of the inactive state of the invention. In the inactive state when a lever handle is parallel to component **496** said valve **632** is normally open while said valve **482** is normally closed. Said leading edge **486** of piston component **491** is substantially close to valve **482** and additionally, outlet spring **483** and inlet spring **621** are in a relaxed state. In FIG. **78a** the leading edge **486** and valve **632** is positioned between the lines **1021** and **1022**. The outlet **478** would be positioned between lines **1022** and **1023**. The outlet **478** diametric dimension is less than that of the separation between lines **1022** and **1023** thus by choosing the travel of the leading edge before opening of valve **478** to be greater than the separation between lines **1021** and **1022** both valves **632** and **482** can be closed to enable build up of pressure in **487** before valve **482** opens to dispel fluid through conduit **478** while valve **632** remains closed.

[0236] It is preferred that detergent be expelled substantially at the same time as the flow of water through spout **10** or immediately prior to the commencement of water flow through spout **10**. Prior expulsion of detergent is enabled when the moveable disc **102** sufficiently blocks area **114** of the stationary disc **112** to enable turning of stem **104** by the lever **20** to enable the action crank **460** and follower **496** to activate detergent expulsion prior to water flow through openings **114** of the stationary disc. Thus, geometric relation between the angular position of the crank **410** on the stem **104**, the lift of said crank **410**, the follower **495**, the leading edge **464** and valve **482** is enabled to allow timely expulsion of detergent within the interval of turning of said lever **20**.

[0237] Referring to FIG. **79** what is shown is a view of an integrated configuration of said crank unit **410** shown in FIG. **39**, said cartridge component **420** shown in FIG. **40**, with a sectional view of said detergent activation unit shown in FIG. **41**, said inlet valve component unit shown in FIG. **42** and said detergent containment unit shown in FIG. **43**. For the configuration of FIG. **79**, when said lever **20** has turned clockwise at a predetermined angle the crank unit **410** is at its maximum displacement and the spring **483** exerts a maximum restoring force on the outlet valve unit **482**. Additionally, said piston **491** is substantially constrained by channel **492** to only motion by translation consequently enabling predictable functioning of the invention.

[0238] Referring to FIG. **79**, the rotation of lever **20** enables beveled edge **488** of said piston **491** to function as a wedge to intercept and initiate vertical translation of valve member **632** thus dislodging inlet valve component **682** shown in FIG. **64** from its seat **629** and thus initiating distortion of spring **621** and thus enabling a restoring force on said spring **621** and consequently enabling member **632** to propagated normal force on said piston unit component **449** shown in FIG. **57**.

[0239] Thus piston unit component **449** shown in FIG. **57** is acted on by an axial force by said outlet spring member **483** and a normal force by inlet spring member **621** thus when said inlet spring member **621** is enabled with a superior spring constant that opposes the influence of spring **483** on said piston unit **449** said piston unit **449** is disposed to be immobilized and consequently disabling any influence of the spring **483** on the rotation stem **104**.

[0240] Additionally, vertical motion of valve member **632** further enables closure of said valve **632** thus disabling flow of liquid detergent through conduits **628** to region **487** of cylinder **464** consequently enabling build up of fluid pressure in region **487** of said cylinder **464** when said lever

**20** continues to rotate and said valve member **632** continues to maintain a stress on said piston member **449** when valve **482** is still closed.

[0241] Referring to FIG. **79**, said lever **20**, said crank unit **460** and said stem **104** substantially function as a rigid body thus clockwise motion of lever **20** enables said crank **410** to translates piston **491** consequently enabling piston extension **486** to intercept and translate and open outlet valve **482** and thus enable ejection of detergent fluid through outlet **478**.

[0242] Referring to FIG. **79**, said crank **460** on its rise interval would have translated said piston extension a measure substantially that of the length of the crank **460**. Thus, spring component **483** would thus have been distorted a displacement substantially that of the length of the crank **460** when said distorted displacement of said spring member **483** is allowable by the physical characteristics of said spring component **483**. Said crank member **460** is thus predetermined to execute its maximum translation prior to the execution of the clockwise maximum predetermined angle of rotation of said lever **20**.

[0243] At the initial turn of lever **20** when detergent has been enabled and water flows through spout **10** hands are enabled to be washed and the faucet counter turned to its off mode without enabling washing of lever **20**.

[0244] Referring to FIG. **80**, what is shown is a horizontal cross-section of the configuration depicted in FIG. **79**.

[0245] Referring to FIG. **80a**, what is shown is the closure of valve **632** and the opening of valve **482** between lines **1022** and **1023**. Crank **460** now is at its maximum lift while spring **482** is now exerting a maximum restoring force simultaneously as spring **621** exerts a force on piston component **491**. The lever **20** has not yet turned through its maximum predetermined angle. The geometric relation of the stationary disc **112**, the moveable disc **102** and the configuration of lever **20** relative to fixed stops **60** and **61** were said fixed stops **60** and **61** are positioned on the upper surface **130** is shown in FIG. **25**.

[0246] Referring to FIG. **81**, what is depicted is a configuration wherein said lever **20** has executed its clockwise maximum predetermined angle of rotation and said lever **20** is deterred from further rotation by said limiting cartridge body stop. Said connecting rod **496** would have retreated with reduced influence of spring **483** with maximum influence of spring **621** on piston **491** with further rotation restricted by body stop **61** and the force of spring **621** acting on piston **491** thus solvent such as water is enabled to flow without interruption when the invention is being utilized. Valves **632** and **483** are both closed for the configuration of FIG. **81**.

[0247] Referring to FIG. **82**, what is depicted is a top view configuration of the configuration shown in FIG. **81**.

[0248] Referring to FIG. **82a** what is shown is a diagrammatic representation in the active state when the lever **20** has been turned through its maximum predetermined angle of turn and the leading edge **464** has retreated from the valve **482** while being acted on by a reduced axial force by spring **483**.

[0249] Referring to FIG. **82a**, the crank **460** has also now retreated and the choice of the dimension of the cross section of outlet conduit **478** and the separation between line **1022** and line **1023** has enabled the closure of valves **482** and valve **632**. The combined force of action by spring **621** on the piston **491** and the reduction of the force of spring **482** coupled with the off set in combination with the action of stop member **61** immobilizes the lever at it maximum predetermined angle of rotation thus water or any other solvent is enabled to flow the spout **10** without interruption for the washing of hands.

[0250] A counter turn of lever **20** enables said lever **20** to intercept the line of flow **777** from said spout **10** thus enabling washing of the lever and further washing of hands. Further counter turn of said lever to its rest position enables refilling of said region **487** of cylinder **464** in preparation for the utility of the invention.

[0251] Referring to FIG. **82b** what is shown is a section in comparison to FIG. **82a** that includes

outlet **478**. When crank **460** retracts at the turning of lever **20** to its maximum predetermined angle outlet **478** is enabled between region **1023** and **1024** thus enabling valve **632** and valve **482** to be in closure mode thus minimizing dripping or leakage.

[0252] When said system depicted in FIG. **81** is being deactivated after use said lever **20** is rotated counter clockwise thus enabling closure of valve **482** while said valve component **632** opens, thus further counter clockwise rotation of lever **20** enables a pressure difference in region **487** and thus enables flow of detergent in said region **487** when said valve member **482** is enabled to be closed prior to the opening of valve **632**. Said system is now in a state for its next utility function.

[0253] The detergent activating unit **430** is disposed to be positioned within a circle of influence with the detergent activation unit **430**, thus enabling flexibility in the positioning of said detergent activation unit **430** on a faucet mount. Additionally multiple configurations are enabled for timely dispensing of fluid while lever **20** is being rotated.

[0254] Referring to FIG. **83** what is shown is a vertical section in a plane symmetric to the detergent activation unit and the detergent reservoir in relation to the position of the crank unit **410**, the cartridge unit and the lever unit in the inactive or closed mode of the invention.

[0255] Referring to FIG. **84**, what is shown is a configuration of the lever, the movable disc **102** and the fixed disc for the closed mode for FIG. **83**.

[0256] Referring to FIG. **85** what is shown is a vertical section in a plane symmetric to the detergent activation unit and the detergent reservoir in relation to the position of the crank unit **410**, the cartridge unit and the lever unit in the active or open mode of the invention.

[0257] Referring to FIG. **86**, what is shown is a configuration of the lever, the moveable disc **102** and the fixed disc in the open mode for FIG. **85**.

[0258] Referring to FIG. **87** what is shown is a sectional configuration of a substantial cylindrically disposed cartridge mount **820** enabled with slots **821** for mounting the cartridge unit **420**. Said member **820** functions to laterally encloses cartridge unit **420** with space between the inner wall of **820** and the outer wall of cartridge body **90** for the cartridge unit **420**. Said member **820** also functions to arrest rotary motion of said cartridge body **90** when said cartridge body **90** is associated with said cartridge mount **820** and said tabs **128** of said cartridge body **90** are recessed in symmetrically arranged slots **821** on the perimeter region **819** of said member **820**. In an assembled unit of the invention cartridge valve unit **420** is stabilized in region **801** of said cartridge mount **820** when flanges **128** of said cartridge body **130** are recessed in slots **821** of said cartridge mount **820**.

[0259] Referring to FIG. **87**, said cylindrically disposed member **820** is configured with a grooved upper end **819** with a smaller diameter for accepting bonnet nut **205**. Said member **820** lower end is configured with a larger diameter region **826**. Said larger diameter region **826** is configured with a spirally grooved access **827** for connection to an inlet conduit and a preferably spirally grooved region **831** for connection to an outlet conduit. The outer surface of said cartridge mount **820** between the upper reduced diameter region **819** and the lower larger diameter region **826** is spirally grooved for rigidly attaching said cartridge mount **820** by nuts to a mounting surface. Such a mounting surface **981** is depicted in FIG. **90**.

[0260] Referring to FIG. **88a**, what is shown is interface **861** for cartridge unit **420**. Said interface **861** includes a conduit **70** for allowing water through the openings **114** of the stationary disc **112**. Surface **859** on upper portion of **861** interfaces with the lower surface **123** of fixed disc **112** by way of a seal **88** to provide a seal between member **861** and said stationary disc **112** to prevent leakage between **861** and **820**. A space **900** is enabled between the inner wall of cartridge mount **820** and the outer perimeter of flange surface **858** of said interface **861** consequently enabling output flow from region **110** of the moveable valve **102** to region **829** of said cartridge mount **820** thus enabling fluid flow through **865** to spout **10**; said **861** is then firmly attached to region **827** of cartridge mount **820** when the fixed disc **112** is situated in region **132** of said cartridge body **90**.

[0261] Referring to **88b** what is shown is an interfacial inlet attachment **863** with conduit **70**. Said interfacial conduit has a larger externally spirally grooved diameter region **866** and a smaller

externally spirally grooved diameter region **869**. Said outer grooved larger diameter region **866** locks into inner grooved diameter region **824** of **861** and preferably soldered to **861**. Means **868** is provided for insertion of said conduit **863** into grooved region **824** of **861**. The lesser diameter region **869** of **863** interfaces with water line attachments from an inlet source. FIG. **88c** shows a section view of said inlet conduit **863**.

[0262] Referring to FIG. **88d** what is shown is an outlet attachment **865** with conduit **874** for connecting to the spout **10** of the faucet unit. Said outlet conduit **865** is configured with externally spirally grooved end portions **870** and **872** and a region **871** between said grooved region **870** and said grooved region **972**. Said region **871** is enabled with means for insertion into access **831** of cartridge mount **820**. It is preferred that **865** be soldered to cartridge mount **820**.

[0263] Referring to FIG. **88e** when said interface **861** is firmly locked in **827** of the lower region **826** of **820** and said lower surface **123** of fixed disc **112** of cartridge **420** interfaces with the upper surface **859** of **861** and said bonnet nut **205** is attached to **820** to allow the flange **47** of said bonnet nut **205** to be firmly applied over the upper surface **130** of the cartridge body **90** and open region **45** of said bonnet nut **205** fully encloses region **59** of the stem **104** then said stem **104** is disposed to be stabilized for interaction with the disc assembly. A washer **822** interfaces **820** and **205**.

[0264] Referring to FIG. **88e** when the bonnet nut **205** is in firm relation with the upper surface **130** of body **90** of the cartridge unit **420** and said bonnet nut **205** is firmly attached to cartridge mount **820** a seal **88** between the upper surface of **861** and the lower surface **123** of fixed disc **112** is included to prevent leakage between the outer surface of **861** and the inner surface of cartridge **820**. The seal **88** interfaces flange **859** and surface **858** of the upper surface of said **861** with the lower face **123** of the fixed cartridge **112**. The lower grooved end **864** of said **861** would also be firmly attached at **827** of said cartridge mount **820**.

[0265] Referring to FIG. **88e** said upper region **858** of **861** is configured to accept an elastomeric seal **116** between a ledge **858** of **861** and the lower part **132** of said cartridge body **90**; such a seal **116** functions to prevent leakage between the outer surface of **861** and the interior surface of **820**.

[0266] Referring to FIG. **88e** the attachment between the lower portion **824** of **861** and the lower portion **826** of **820** is enabled with an elastomeric seal **119**. Said seal **119** functions to prevent leakage between **861** and **820**.

[0267] Referring to FIG. **89** what is shown is the embodiment of FIG. **88e** when input mains water supply attachment **863** is attached and preferably soldered to **861** at access opening **824** of said **861** and output water attachment **965** is attached and preferably soldered to **820** at access opening **831** of said cartridge mount **820**. Said conduit **865** functions to direct water to spout **10** when a water line is connected from conduit **865** to spout **10**. Said attachments **865** and **863** are preferably made from copper, brass or stainless steel and soldered to member **820**.

[0268] Referring to FIG. **90** what is shown is a section of embodiment **981** for mounting the invention for use with a wash basin. To secure the detergent activation unit **430** shown in FIG. **48** more ways than one is available; one available means for securing said detergent activation unit involve having access slots **935** for accepting fasteners to which is attached a nut for bonding said detergent activation unit **430** to said embodiment **981**. Said embodiment member **981** has access openings **947** for receiving cartridge housing **820** for enabling nuts **933** to firmly secure said cartridge housing **820** to said mounting base **981**. Said embodiment member **981** has access openings **947** for receiving cartridge housing **820** to allow nuts **933** to firmly secure said cartridge housing **820** to said embodiment **981**.

[0269] Referring to FIG. **90**, said embodiment **981** is enabled with access **939** for receiving spout assembly to allow a nut to secure said spout assembly to said base **981**.

[0270] Referring to FIG. **90**, said mounting base **981** is enabled with opening **943** for setting detergent activation unit **430** and also with access openings **935** for receiving bolts **948**. Said bolts **948** firmly associate undercover plate **930** and detergent activation unit to **430** when a nut **942** is applied to **948** in such a manner that the members **930**, **981** and base member **465** of the cartridge

activation unit **430** are contiguous and firmly associated.

[0271] Referring to FIG. **91** what is shown is the embodiment of an under-cover plate **930** for interfacing with mounting base **981** when said **981** is in relation to a wash basin. To secure the under-cover plate **930** more ways than one is available; one available means for securing said detergent activation unit involve having access slots **935"** for accepting bolts **948** to which is attached a nut **942** for securing said detergent activation unit **430**. Said embodiment member **930** has access openings **947"** where the diameter measurement of said openings **947"** is greater than the measurement of the most significant diameter of nuts **933**. Said openings **947"** enable access for cartridge mount **820**.

[0272] Referring to FIG. **91** said embodiment **930** is enabled with access **939"** for receiving spout assembly **10** and to allow a nut **933** to secure said spout assembly **10** to said base **981** and to said under-cover plate **930**.

[0273] Referring to FIG. **91**, under-cover plate **930** is enabled with opening **943"** for setting detergent activation unit **430** and also with access openings **935"** for receiving bolts **948**. Said bolts **948** firmly associate undercover plate **930** and detergent activation unit to **930** when a nut **942** is applied to **948** in such a manner that the members **930**, **981** and base member **465** of the cartridge activation unit **430** are contiguous and firmly associated.

[0274] FIG. **92** what is shown is an escutcheon embodiment **940** for mounting on surface **981** when said embodiment is in relation to a wash basin. To secure the escutcheon **940** more ways than one are available; one available means for securing said escutcheon **940** involve having access slots **955** for accepting lever assembly **304** configured so that when said lever assembly is attached to region **58** of stem portion **57** of stem **104** the lower surface of said lever assembly abuts the surface of escutcheon embodiment **940**. To further secure the escutcheon **940**, or protective covering **940**, when spout **10** is included in opening **953** and escutcheon **975** is attached to the lower end of said spout **10** the lower edge of escutcheon **975** abuts the surface of escutcheon **940**.

[0275] Said escutcheon **940** shown in FIG. **92** is also enabled with access opening **957** where said access opening **957** accepts lower portion **724** of detergent reservoir unit **450** when said lower portion **724** of the said detergent reservoir unit **450** is attached to region **487** of the detergent activation unit **430**, a nut for securing said detergent activation unit **430**. Referring to FIG. **92** access opening **951** enables availability of detergent when outlet conduit **478** is accepted by said opening **951** of said detergent activation unit **430**.

[0276] Referring to FIG. **92a** what is shown is a spout assembly **975** for inclusion with spout **10** where said spout escutcheon **977** of spout assembly **975** is enabled to abut an elastomeric washer **976** and said elastomeric washer **976** abuts surface **981** in an assembly of the invention and is in a relation to the lower region of said spout **10**. Nut **944** is applied to the under surface of **981** to firmly attached spout assembly **975** to said **981**. An elastomeric gasket **945** is preferred between said nut **944** and the under surface of **981** in the assembly of the spout assembly **975**. Opening **939"** of undercover gasket **930** is enabled to accommodate the diametric dimensions of nut **944**.

[0277] Referring to FIG. **93** what is shown is an explosion of nut **933**, rubber washer **934** and cartridge housing **820**, the detergent activation unit **430**, the fasteners **948** and nuts **94** in relation to mounting base **981**.

[0278] FIG. **93a** shows a front view of said nut **933**, said washer **934** and said cartridge housing **920** assembled on faucet mounting base **981** shown in FIG. **93**.

[0279] Referring to FIG. **94** what is shown is the spatial positioning of order of assembly of detergent activation unit **430**, crank unit **410** and cartridge unit **430**, bonnet nut **205** and stop attachment **404** for an embodiment in which the detergent activation unit **430** is activated when a right hand-side lever **20** is rotated. The lever handle **305** is not shown.

[0280] Referring to FIG. **94a** shows the assembly of FIG. **94** with escutcheon **940**.

[0281] Referring to FIG. **94b** shows the assembly of FIG. **94a** with spout assembly **975** and lever handle **304**. Screws for attaching lever unit **304** and escutcheon for said screws are not shown.

[0282] Referring to FIG. 95, what is shown is the configuration of FIG. 94b assembled with spout unit **10**, lift rod **18** and bonnet nut **205** were said bonnet nut **205** is firmly associated with cartridge mount **820**. When said bonnet nut **205** is firmly associated with cartridge mount **820** and region **45** of said bonnet nut **205** surrounds region **59** of stem assembly **104** optimum stability of said stem **104** is facilitated. The outer surface of region **59** of stem unit **104** is enabled to rotate substantially frictionally against the inner surface **45** of the bonnet nut by the application of a Teflon coating to said surface **59** of stem unit **104**.

[0283] Referring to FIG. 96, what is shown is the configuration of FIG. 95 assembled with the detergent reservoir unit **450** in region **487**. Said access opening **487** is spirally grooved to be compatible with the externally spirally grooved portion **724** of reservoir unit **450** such that when portion **724** is spiraled into access **487** said detergent reservoir **450** is firmly attached to the detergent activation unit **430**.

[0284] Referring to FIG. 97, what is shown is an exploded conceptual view of an embodiment of the invention with under cover plate **930**. The diameter of openings **947** of said undercover plate **930** is superior to that of the most significant diameter of nut **933** thus minimizing interaction influence with the cartridge unit when the detergent activation unit is being activated.

[0285] Additionally, said under cover plate **930** interfaces securely with mounting **981** when nut **944** is attached to region **955** of spout **10** shown in FIG. 9; in such a manner said nut **944** functions to secure **930** to the lower face **981**. Said under-cover plate is further stabilized when screws **948** are introduced from the upper surface of **498** through openings **497** of said **498** and then through openings **935** of **981** and then through openings **935"** of **930** thus enabling screw fasteners to be chosen and be applied to firmly bond together the body **465** the of detergent activation unit **430**, the mounting block **981** and said undercover plate **930** by one or more screw nut fasteners **942**.

[0286] As such and in summary, the figures depict a washable faucet lever with crank-activated detergent dispenser having a lever cartridge **420** having a stem **104** operably coupled to a hand lever **20** configured to rotate in a lever axis of rotation and operably coupled to a two-bar linkage configured to rotate in a linkage axis of rotation parallel to and simultaneous with the lever axis of rotation. Said another way, the piston **491** is configured to translate in a single and linear axis of translation within the enclosed region **468** upon rotating the hand lever **20** in the lever axis of rotation to selectively and simultaneously displace the reservoir valve **632** and outlet valve **482**. Stated differently, the piston **491** is configured to translate within the enclosed region **468** upon rotating the hand lever **20** in the lever axis of rotation and in either a counterclockwise or clockwise rotation to selectively and simultaneously displace the reservoir valve **632** and outlet valve **482**, i.e., one impetus or rotation of the hand lever **20** will cause the valves **632**, **482** to open (or close).

[0287] The washable faucet lever with crank-activated detergent dispenser also includes a detergent reservoir member **450** configured to house a liquid detergent therein, having an internal channel **718** configured to receive the liquid detergent, and having a reservoir valve **632** coupled to the detergent reservoir member **450** and operably configured to selectively translate to open and permit flow of the liquid detergent through the internal channel **718** and to close and restrict flow of the liquid detergent through the internal channel **718**. Further, the dispenser also includes an encasement member **498** defining an enclosed region **468** therein, defining an outlet **478**, having an outlet valve **482** disposed within the enclosed region **468** and operably configured to selectively translate to open and permit flow of the liquid detergent through the outlet **478**, to close and restrict flow of the liquid detergent through the outlet **478**, and biased with a spring **483** retained by the encasement member **498**.

[0288] Further, the washable faucet lever with crank-activated detergent dispenser may also include a piston **491** disposed within the enclosed region **468**, coupled to the two-bar linkage, and operably configured to translate within the enclosed region **468** upon rotating the hand lever **20** in the lever axis of rotation to selectively and simultaneously displace the reservoir valve **632** and outlet valve **482** to permit flow of the liquid detergent through the internal channel **718**, into the enclosed region

**468**, and out through the outlet **478**. In one embodiment, the linkage axis of rotation is coaxial with respect to the lever axis of rotation.

[0289] In one embodiment of the present invention a mounting base **981** is utilized and that defines an opening with the lever cartridge **420** disposed therein and coupled thereto with at least one fastener, coupled to the encasement member **498** with at least one fastener, and defines another opening with a faucet member disposed therein. The faucet member includes a spout **10** with an opening configured to emit liquid therethrough and is coupled (i.e., directly or indirectly) to the mounting base **981** with at least one fastener. The dispenser will also include a protective covering **940** defining an opening with the lever cartridge **420** disposed therein, covering the encasement member **498**, defining an opening with the faucet member disposed therein, and defining an opening with the detergent reservoir member **450** disposed therein.

[0290] In some embodiments, the protective covering **940** defines an internal channel fluidly coupled to the outlet **478** and configured to permit flow of the liquid detergent therethrough and the reservoir valve **632** is biasedly coupled to the detergent reservoir member **450** with a spring. Also in some embodiments, the internal channel **718** is spatially configured in the detergent reservoir member **450** to enable gravity-based receipt of the liquid detergent therein.

[0291] Referring to FIGS. **98a**, **98b** and **98c**, what is shown is another embodiment of the water line system into the cartridge unit and water line system out of the cartridge unit for the invention. The embodiment shown for FIG. **98a**, FIG. **98b** and FIG. **98c** are an alternative for what has been previously described and thus only information related to this embodiment for installation and assembly needs to be described. The valve chamber is composed of an inlet tube **961'**, a main tube body **920'** and an outlet/inlet sleeve **963'**. The inlet tube **961'** is preferably made from copper of brass or stainless steel and formed with an inlet tunnel **70'**. The outer circumference of the upper end of the inlet tube **961'** is formed with an annular engaging section **958'** with a larger diameter. The middle portion is milled to form a stepped section. An annular flow guiding rib **967'** with a larger diameter is adjacent to the lower side of the stepped section. A water scaling ring **119'** is fitted under the flow guiding rib **967'**. The lower end of the inlet tube **961'** under the flow guiding rib **967'** is milled with a locking outer thread section **964'**. The main tube body **920'** is also made from copper, brass or stainless steel and milled to form an internal receiving tunnel **901'**. The inner circumference of the receiving tunnel **901'** near the lower end thereof is formed with an annular stop section **925'**. The outer circumference of the upper end of the main tube body **920'** is formed with an outer thread section **921'** with a smaller diameter. The edge of the upper opening is cut with two diametric notches **921'**. The middle portion of the main tube body **920'** is formed with a main outer thread section **903'** with a certain length. The lower end of the main tube body **920'** under the main thread section **903'** is a smaller diameter section milled with fastening outer thread section **902'**. A water sealing ring **117'** is fitted on the small diameter section. The outlet/inlet sleeve **963'** is made by pressing cast. The center thereof is with a stepped adapting flow passage **910**. The opening of the larger diameter section of the adapting flow passage **910'** is formed with a fastening inner thread section **904'** and the smaller diameter section thereof is formed with a locking inner thread section **909'**. A lower side of the outlet/inlet tube **50'** is formed with an inlet tube **908'** and a lateral side thereof is formed with an outlet tube **906'**. The inlet tube **908'** and the outlet tube **906'** both communicate with the adapting flow passage **910'**. In addition, the outer circumference of the end of the inlet tube **908'** is formed with an inlet outer thread section **907'** while the outer circumference of the end of the tube **906'** is formed with an outlet outer thread section **905'**.

[0292] Referring to FIG. **99**, when assembled the inlet tube **961'** is first fitted with the receiving tunnel **901'** of the main tube body **920'** with the engaging section **958'** engaged with stop section **925'** of main tube body **920'**. At this time, the inlet tube **961'** is preliminary assembled with the main tube body **920'** with locking outer thread section **964'** of the inlet tube **961'** extending out of the main tube body **920'** to be screwed in with the locking inner thread section **909'** of the outlet/inlet sleeve **963'**. The locking outer thread section **47' 902'** of the main tube body **920'** is

screwed with the fastening inner thread section **904'** of the outlet/inlet tube **963'**. Under such circumstances the inlet tube **961'**, main tube body **920'** and the outlet/inlet tube **963'** are integrally associated together. A valve chamber **801'** with a certain depth is defined between the upper end of the engaging section **958'** of the inlet tube **961'** and the upper opening of the receiving tunnel **901'** of the main tube body **920'**. Also, an annular outlet passage **900'** with a certain size is defined between the outer circumference wall of the tube **961'** and the inner circumference walls of the main tube body **920'** and the large diameter section of the outlet/inlet sleeve **963'**. At this moment, the insertion of the cartridge unit and stabilization of the cartridge by bonnet nut completes the assembly for the functioning of the cartridge chamber.

[0293] Referring to FIG. **100**, what is shown is an alternative embodiment of the stem portion **57'** for the cartridge **604**. Said stem portion **57'** has an upper region **258** configured as a rectangular prism for inserting said portion **258** into rectangular access **234** of the lever assembly **814**; said lever assembly **814** is shown in FIG. **103**. Contiguous with portion **258** of stem portion **258** is cam attachment region **252**; said region **252** fits into rectangular access opening **234** of cam unit **810**. Crank unit **810** is shown in FIG. **106**. Contiguous with region **252** of stem portion **57'** is a cylindrical region **259** where said cylindrical region **259** fits into smooth inner circular opening **45** of bonnet nut **205** shown in FIG. **105**. Said bonnet nut **205** when assembled is firmly attached to cartridge housing **920** and thus said bonnet nut is disabled from motion when the lever assembly **814** is rotated. A section view of bonnet nut **205** is shown in FIG. **104**. Referring to FIG. **101**, what is shown is an alternative embodiment **89** for intercepting body stops **260** and **261** and thus restricting the rotation of lever unit to its maximum predetermined angle of rotation. Said embodiment **89** has rectangular opening that intimately fits over region **256** of region **57'** of the stem unit **104'** shown in FIG. **100**. FIG. **102** shows a configuration of **89** fitted onto the upper surface **230** of the cartridge body **90**.

[0294] Referring to FIG. **107**, when the cartridge stem is already inserted in body passage **68** member **89** is first attached to region **256**; bonnet nut **814** is then applied to the cartridge housing **920** to stabilize the cartridge unit; bonnet nut **814** now completely encloses member **89**. Subsequently cam body unit **810** is attached above the bonnet nut and followed by attachment of lever assembly **814**.

[0295] Other features and advantages will become apparent from the following description preferably taken relative to the accompanying drawings, which in combination discloses an embodiment of the invention. For example, FIG. **108** shows an alternative embodiment for interacting with the detergent activation unit.

## Claims

**1.** A washable faucet lever with crank-activated detergent dispenser comprising: a lever cartridge having a stem operably coupled to a hand lever configured to rotate in a lever axis of rotation and operably coupled to a two-bar linkage configured to rotate in a linkage axis of rotation parallel to and simultaneous with the lever axis of rotation; a detergent reservoir member configured to house a liquid detergent therein, having an internal channel configured to receive the liquid detergent, and having a reservoir valve coupled to the detergent reservoir member and operably configured to selectively translate to open and permit flow of the liquid detergent through the internal channel and to close and restrict flow of the liquid detergent through the internal channel; an encasement member defining an enclosed region therein, defining an outlet, having an outlet valve disposed within the enclosed region and operably configured to selectively translate to open and permit flow of the liquid detergent through the outlet, to close and restrict flow of the liquid detergent through the outlet, and biased with a spring retained by the encasement member; and a piston disposed within the enclosed region, coupled to the two-bar linkage, and operably configured to translate within the enclosed region upon rotating the hand lever in the lever axis of rotation to selectively

and simultaneously displace the reservoir valve and outlet valve to permit flow of the liquid detergent through the internal channel, into the enclosed region, and out through the outlet.

**2.** The washable faucet lever with crank-activated detergent dispenser according to claim 1, wherein the piston is configured to translate in a single and linear axis of translation within the enclosed region upon rotating the hand lever in the lever axis of rotation to selectively and simultaneously displace the reservoir valve and the outlet valve.

**3.** The washable faucet lever with crank-activated detergent dispenser according to claim 1, wherein the piston is configured to translate within the enclosed region upon rotating the hand lever in the lever axis of rotation and in one of a counterclockwise or clockwise rotation to selectively and simultaneously displace the reservoir valve and the outlet valve.

**4.** The washable faucet lever with crank-activated detergent dispenser according to claim 1, wherein the linkage axis of rotation is coaxial with respect to the lever axis of rotation.

**5.** The washable faucet lever with crank-activated detergent dispenser according to claim 1, further comprising: a mounting base defining an opening with the lever cartridge disposed therein and coupled thereto with at least one fastener, coupled to the encasement member with at least one fastener, and defining an opening with a faucet member disposed therein, having a spout with an opening configured to emit liquid therethrough, and coupled thereto with at least one fastener; and a protective covering defining an opening with the lever cartridge disposed therein, covering the encasement member, defining an opening with the faucet member disposed therein, and defining an opening with the detergent reservoir member disposed therein.

**6.** The washable faucet lever with crank-activated detergent dispenser according to claim 5, wherein the protective covering defines an internal channel fluidly coupled to the outlet and configured to permit flow of the liquid detergent therethrough.

**7.** The washable faucet lever with crank-activated detergent dispenser according to claim 1, wherein the reservoir valve is biasedly coupled to the detergent reservoir member with a spring.

**8.** The washable faucet lever with crank-activated detergent dispenser according to claim 1, wherein the internal channel is spatially configured in the detergent reservoir member to enable gravity-based receipt of the liquid detergent therein.

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